Better Predictions, Better Decisions? Experimental Evidence on Algorithmic Tracking across Seven Colleges

Peter Bergman[†], Elizabeth Kopko[§], Julio Rodríguez[‡]

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ABSTRACT

Many consequential allocation decisions are prediction problems, but two obstacles stand between better predictions and better decisions. First, the data used to evaluate predictions are often generated under the incumbent assignment rule, which induces a selection problem that could increase prediction error. Second, predictive gains are realized only through the thresholds institutions choose under real constraints. We study these issues in educational tracking: 71 percent of U.S. colleges use test scores to track students into courses. Inaccurate placement has consequences: students face misaligned curriculum and must pay tuition for remedial courses that do not bear credits toward graduation. We develop an alternative system that uses algorithms to predict college readiness and track students into courses. Compared to the most widely-used placement tests in the country, the algorithms are more predictive of future performance. We conduct a natural field experiment across seven colleges to evaluate the effects of algorithmic placement. Placement rates into college-level courses increase substantially without reducing pass rates. Algorithmic placement generally, though not always, narrows differences in college placement rates and remedial course taking across demographic groups. We use the experimental design and variation in placement rates to assess bias in each system. Test scores exhibit substantially more discrimination than algorithms; a significant share of test-score disparities between Hispanic or Black students and white students is explained by discrimination. We also show that the selective labels problem nearly doubles the prediction error for college English performance but has almost no impact on the prediction error for college math performance. A detailed cost analysis shows that algorithmic placement increases college credits earned at a modest net cost to the government, implying a marginal value of public funds of at least 1.1.

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[†] University of Texas at Austin and NBER.
[§] Community College Research Center.
[‡] University of Oxford.

1. Introduction

Many consequential allocation decisions are prediction problems (Kleinberg et al., 2015). Institutions decide who receives credit, medical treatment, employment, or access to educational opportunities by forecasting future outcomes. Better predictions can improve these decisions, but implementation is challenging. First, the data used to evaluate predictions are often generated under the incumbent assignment process; this selection problem makes it difficult to evaluate how accurate predictions would be under a new assignment rule. Second, the value of a better prediction is only realized through the rule it supports: institutions choose thresholds and implement rules under constraints that can erode predictive gains, with consequences that differ across groups.

We study algorithms in one such allocation decision: educational tracking. Each year, U.S. colleges decide which of millions of entering students are ready for college-level coursework. At least 71 percent of postsecondary institutions use test scores to track students into college-level coursework or remediation (Fields and Parsad, 2012).[1] The share is higher at two-year colleges, which enroll nearly half of all postsecondary students while fewer than 40 percent of these students receive any credential (Bound et al., 2010; Denning et al., 2022). While there are large potential benefits of tracking (cf. Duflo et al., 2011; Banerjee et al., 2016; Banerjee et al., 2017; Banerji and Chavan, 2016; Card and Giuliano, 2016), *inaccurate* placement has consequences: students face misaligned curriculum, and, in higher education, students must pay tuition for remedial courses that do not bear credits toward graduation. The potential for inaccurate placement is a concern because there is evidence that several widely-used tests lack validity and unnecessarily reduce educational opportunities for students from under-represented groups (Rothstein, 2004; Scott-Clayton et al., 2014).

We develop placement algorithms to track students into college-level or remedial courses and implement these algorithms via a natural field experiment across seven colleges and 12,526 college students (Harrison and List, 2004; Levitt and List, 2009; List, 2011). We recruited community colleges across New York and gathered historical data on their students to estimate models predicting students' likelihood of passing college-level math and English courses. These predictions incorporated measures such as placement-exam scores, high school GPA, high school rank, diploma status, and time since high school graduation. We created placement algorithms for math and English that placed students into a remedial course if a student's predicted probability of passing a college-level course was below a cut point chosen by each college. We then randomly assigned

[1] Outside the U.S., test scores are frequently used as the sole criterion for admission to colleges (Aguirre et al., 2020; Hastings et al., 2013; Kirkeboen et al., 2016; MacLeod et al., 2017; Riehl, 2019).

students to either colleges' test-score placement system—from the most widely-used company in the country—or the placement algorithms.

A key contribution of our paper is the experiment. First, relying on historical data and placement rules faces a "selective labels problem" (Kleinberg et al., 2018; Lakkaraju et al., 2017). Though selection into college-level courses is based on observables, we can only estimate and evaluate the algorithm conditional on placement into the college-level course. The algorithm recommends placement into the college-level for some students who never would have been placed as such before. At best, such recommendations rely on extrapolations that may be inaccurate. Another challenge is that colleges require rapid placement results and integration into their existing systems. This speed requirement limits the complexity of the algorithms we could implement. Experimental evaluation of algorithmic placement is critical for testing how well it performs in practice.

Second, the experiment provides variation in placement rates that helps us address the selective labels problem. In typical observational settings, course outcomes are observed among students who are placed into college-level courses, which means predictors are estimated on a selected sample. By randomly assigning students to two systems with different placement decisions, we observe performance in college-level courses for a much broader set of students, including many who would never have reached college-level courses under a single system. This variation is valuable because it allows us to evaluate the algorithm more completely: we measure bias in each placement system (Baron et al., 2024a; Baron et al., 2024b), and we show how selective labels biases our measures of the algorithm's accuracy.

Lastly, the experiment allows us to provide suggestive evidence on whether an algorithm that places students according to who benefits most from college or remedial courses would place students differently. To do so, we use causal forests (Wager and Athey, 2018) combined with random assignment and variation in placement. We then measure how placement rates would change if we assigned students to college-level versus remedial coursework based on which course they would benefit from most.

The impacts of algorithmic placement systems depend on several factors. Improving the validity of the placement instrument could help place students into courses better aligned to their incoming skills. Measures such as high school GPA reflect a wider array of cognitive and non-cognitive skills than test scores alone (Borghans, et al., 2016; Kautz, et al., 2014; Kautz and Zanoni, 2014).[2] The algorithms also help colleges choose cut points for placements into the college-level courses, which affect the number of students placed into these courses and their expected pass rates conditional on placement. This choice means algorithmic placement does not necessarily imply that placement rates will change

[2] GPA also has a high degree of reliability, but there are concerns that grading standards are too school-specific for it to be useful (Bacon and Bean, 2006).

either on net or for a given individual. At particular thresholds (e.g., the extremes), it is possible that the placements assigned by the algorithm and test scores will be the same. But if colleges choose to maintain pass rates, the algorithm may place more students into college-level courses, which could increase students' credit accumulation and save students money. If the algorithm's predictions are sufficiently accurate, it can push colleges toward the efficient frontier tradeoff between pass rates and placement rates.

We find that colleges choose cut points to hold predicted pass rates constant. Colleges define the predicted pass rate as the expected proportion of students earning a grade of C or better based on historical data. This results in large changes in placement rates: relative to the test-score placement system, 23 percent of math placements change and 55 percent of English placements change. Compared to what would have occurred using the test-score placements, the algorithm places 15 percent into a higher-level math class and 49 percent into a higher-level English course. Hence the algorithm places 8 percent of students into a lower-level math course and 6 percent into a lower-level English course.

Placement via the algorithm leads to immediate increases in enrollment into college-level courses. Algorithmic placement yields first-term enrollment increases in college-level math by 2.6 percentage points and in college-level English by 13.6 percentage points relative to the control group. The larger effects in English reflect the lower predictive validity of test scores for English performance relative to math. Algorithmic placement also leads to reductions in remedial course taking and increases in college credits earned *without* reducing pass rates. Placement via the algorithm reduces remedial credits attempted by 1.1 credits and increased college credits earned by 0.56 credits. For students tracked in both math and English the number of college credits earned increases by 1.3 because of algorithmic placement.[3] In the longer run, the credit gains do not reverse, but algorithmic placement does not raise the probability of earning at least 60 college-level credits, which is the minimum credit requirement for an associate degree. For context, this credit accumulation effect is similar in magnitude to the effect of instructor incentives for student performance (Brownback and Sadoff, 2020) for two courses, and slightly less than half the two-semester impact of the comprehensive set of supports offered by CUNY's Accelerated Study in Associate Programs (Weiss et al., 2019). The former did not impact graduation rates, while the latter had large impacts on completion rates in the short and long run.

When the two placement mechanisms disagree on college placement, the algorithm is substantially more accurate. We observe counterfactual placements for all students. When the algorithm recommends placement into college-level math, but the test-score system

[3] Each college has a small number of automatic exemptions from taking a placement exam for a given subject, and, because our placement mechanism was integrated within the testing platforms, not all students could be placed by the algorithm for both math and English.

does not, pass rates are 10 percentage points *higher* than when the test-score system recommends placement into college-level math, but the algorithm does not. For the analogous disagreement in English, pass rates are 12 percentage points higher.

We find evidence algorithmic placement narrows certain demographic gaps in placement rates. The algorithm increases placement into college-level courses for all subgroups we looked at. After adjusting for multiple-hypothesis-testing, increases are significantly larger for female students in math relative to male students and Black students in English relative to white students. Lower-income students have a larger reduction of remedial credits relative to higher-income students. However, the increases in placement into college-level math, though positive and significant, are not as large for Hispanic students compared to white students. Despite increases in placement rates, we find that pass rates in college-level courses remain extremely similar to pass rates in the test-score placement system across different subgroups.

We find the test-score placement system results in significant bias against several under-represented groups, and far more so than algorithmic scores. In math, Black and Hispanic students equally likely to succeed as White students are 15 and 14 percentage points less likely to be placed into college-level math under the test-score system, accounting for 80 and 87 percent of the raw placement gaps, respectively. Disparities are similarly large in English. The algorithm substantially reduces these gaps and for most groups the estimated bias is small and statistically insignificant. We further show that the selective labels problem nearly doubles the prediction error for college English performance but has almost no effect on the prediction error for college math performance.

We find suggestive evidence that an algorithm placing students according to who benefits most from either the remedial or college course in each subject would place even *more* students into college-level courses. While many colleges have moved to co-requisites (Ryu et al., 2022) which allow students to take the remedial course and college-level course simultaneously, our findings suggest that movements to substantially redesign remediation coursework (cf. Schudde and Keisler, 2019) would benefit students more than the status quo and reduce the opportunity costs of co-requisite requirements.

Lastly, the algorithmic placement system reduces the cost of remediation for students. We conducted a detailed cost analysis for colleges and students, separating fixed and variable costs, and costs to students versus costs to colleges. Students saved $220 in remedial tuition, on average, relative to the test-score placement system, and chose to spend $260 more on induced college-level credits, which count toward a degree.[4] The remedial savings amount to $212,960 per cohort, per college, in our sample.

[4] This figure captures only direct tuition savings. Students also avoid the opportunity cost of time spent in unnecessary remedial courses, including delayed graduation and foregone earnings, which likely represents the largest cost of misplacement.

For colleges to implement such a placement algorithm, decision makers must weigh the potential benefits to students against the costs to the colleges. We estimate that the cost per student in the initial year of the study—above and beyond the test-score placement system—is $140. Much of these costs are driven by the need to hand enter data from high school transcripts. Simple process enhancements, such as requesting GPAs on applications, or data-transfer relationships with high schools, could simply and dramatically reduce the cost of this data collection. The first year of implementation also involves large, fixed costs. We estimate operating costs of the placement algorithm are $40 per student. Weighing students' savings against the net cost to the government yields a Marginal Value of Public Funds of at least 1.1 (Hendren and Sprung-Keyser, 2020).

Our paper relates to research arguing that data-driven algorithms can improve human decision making and reduce biases, and do so at low cost (Arnold et al., 2021; Arnold et al., 2022; Li et al., 2020; Ludwig et al., 2024; Mullainathan and Spiess, 2017). Kleinberg et al. (2018) show that a machine-learning algorithm has the potential to reduce bias in bail decisions relative to judges' decisions alone. At the same time, others are concerned that these algorithms could embed biases into decision making and exacerbate inequalities (Eubanks, 2018). We contribute to this literature by comparing the impacts of a simple, data-driven algorithm to another quantitative measure, test scores and evaluating the algorithm by conducting a large-scale experiment.

Our paper also relates to a broader literature on tracking students. Historically, tracking is controversial. Oakes (1985) argues that the evidence on tracking is inconsistent, and, in practice, higher-track classes tend to have higher-quality classroom experiences than lower-track classes. More recently, Duflo et al. (2011) randomize students in Kenya to schools that either tracked students by test scores or assigned students randomly to classrooms. They find that test scores in schools with tracking improved relative to the control group both for students placed in the higher-scoring and the lower-scoring tracks. Card and Giuliano (2016) study a district policy in which students are placed into classrooms based on their test scores. This program caused large increases in the test scores of Black and Hispanic students.

Multiple studies look at the effects of being placed into a higher track versus a lower track. Bui et al., (2014) and Card and Giuliano (2014) find that gifted students' placement into advanced coursework does not change test scores. Cohodes (2020) and Chan (2020), however, find increases in enrollment in advanced high-school coursework and college.[5] In higher education, there is evidence of negative effects on course completion and graduation for marginally admitted students in high-ability classes (de Roux and Riehl, 2022). A

[5] Several other studies look at the effects of placing into high-test score schools and the results are much more mixed (Jackson, 2010; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu et al., 2014; Dobbie and Fryer, 2014).

student's rank within the classroom also shapes performance and major choice (Elsner et al., 2021). On the other hand, the evidence that placement into remedial courses improves academic outcomes for marginal students is more mixed, and several regression-discontinuity analyses find no effects (Allen and Dadgar, 2012; Bettinger and Long, 2009; Boatman and Long, 2010; Calcagno and Long, 2008; Hodara, 2012; Martorell and McFarlin, 2011; Scott-Clayton and Rodriguez, 2015).

The rest of our paper proceeds as follows. Section 2 provides further background information about tracking in postsecondary institutions and study implementation. Section 3 describes the experimental design, data and empirical strategies. Section 4 presents our main findings. Section 5 investigates measures of equity, fairness, and the implications of the selective labels problem for predicting course outcomes. Section 6 provides a detailed cost analysis, and Section 7 concludes.

2. Background, Site Recruitment, Implementation

Tracking students into remedial education is a major component of the higher education system, both in terms of enrollment and cost. In the 2011-12 academic year, 41 percent of first and second-year students at four-year institutions had taken a remedial course, while at two-year institutions, even more—68 percent of students—had taken a remedial course (Chen, 2016). The cost of remedial education has been estimated to be as much as $2.9 billion annually (Strong American Schools, 2008).

Like other programs (e.g. Teaching at the Right Level), the primary purpose of offering remedial education is to differentiate instruction to less-prepared students, so they have the skills to succeed in college-level coursework (Bettinger and Long, 2009). However, there is evidence that community-colleges' tracking systems frequently "under place" students—tracking them into remedial courses when they could have succeeded in college-level courses—and "over place" students—tracking them into college-level courses when they were unlikely to be successful (Belfield and Crosta, 2012; Scott-Clayton, 2012).

At many colleges, incoming students are typically required to take the ACCUPLACER placement exams in math and English unless they qualify for exemptions based on other standardized test scores or prior coursework. Students placed into remedial math or English are strongly encouraged, though not formally required, to enroll in those courses during their first term. However, in our setting, students must take the remedial class prior to the associated college-level course. Placement into college-level courses does not entail a requirement to enroll in those subjects immediately; students may prioritize other college-level coursework based on their program, availability, or advising. Students may also take non-math and non-English college-level courses simultaneously with remedial

classes.[6] As a result, placement determines the recommended entry point into the curriculum but does not guarantee that a student will enroll in the corresponding course.[7] This distinction is important for interpretation: placement affects access, whereas enrollment reflects students' course-taking decisions after placement.

Most institutions administer a multiple-choice test in mathematics, reading, and writing to determine whether incoming students should be placed into remedial or college-level courses. The ACCUPLACER, a computer-adaptive test offered by the College Board, is the most widely-used college placement system in the U.S. (Barnett and Reddy, 2017). Colleges choose a cut score (i.e. a threshold) for each test and place students scoring above this score into college-level courses and students below the cut score into various remedial courses.[8] Given the placement rules and immediate test results provided by the ACCUPLACER platform, students often learn their placement immediately after completing their exam.

Site Selection and Descriptions

All the participating colleges are part of the State University of New York (SUNY) system, ranging from large to small, and students' backgrounds vary from college to college. Table A.1 of the Appendix provides each college's name and an overview of their characteristics using public data. The smallest of the colleges serves roughly 5,500 students while the largest serves over 22,000 students. As is common in community college settings, a large share of the student body is part-time and many are adult learners, with between 21 percent and 30 percent of students over the age of 25. For most of the colleges, the majority of students receive financial aid. The colleges have similar transfer-out rates of between 18 percent and 22 percent and three-year graduation rates are between 15 percent and 29 percent. The colleges also tend to serve local student populations. Lastly, all of the colleges have an open-door admissions policy. This means that the colleges do not have admission requirements beyond having graduated from high school or earned a GED.

[6] Students who fail a college-level or remedial course are generally permitted, and often required, to re-enroll if the course is needed for their program of study. Each attempt counts as a separate registration, and students are responsible for tuition or financial aid usage on every attempt.

[7] As shown in the Results section, enrollment rates are, as expected, lower than placement rates in both subjects.

[8] Certain colleges may offer exemptions from testing. For instance, this can occur for students who speak English as a second language or who have high SAT scores.

Creating the Placement Algorithm

Colleges preferred that we develop college-specific algorithms. We created separate algorithms for each college in math and English using data on each college's previous cohorts of students.

Five colleges in the study had been using ACCUPLACER for several years. One college had been using ACCUPLACER assessments for English but had transitioned from a home-grown math assessment to the ACCUPLACER math assessments too recently to generate historical data, so we tested an algorithm for English placement only at that college. One college in our sample had been using the COMPASS exam, which was discontinued by ACT shortly after this study began. The college replaced the COMPASS exam with the ACCUPLACER exam. At this college, we tested an algorithm that did not use any placement test scores against a placement system that incorporates only ACCUPLACER test results.

We worked with administrators at each college to obtain the data needed to estimate each algorithm. In some cases, these measures were stored in college databases. In other instances, colleges maintained records of high school transcripts as digital images. For the latter, we had the data entered into databases by hand.

To estimate the relationships between predictors in the dataset and performance in initial college-level courses, we restricted the historical data to students who took placement tests and who enrolled in a college-level course without first taking a remedial course. This set of students constituted our estimation sample for developing the algorithm. Importantly, students were selected into college-level courses based on observable characteristics, but this sampling scheme does raise concerns about whether the relationships we estimate between variables will apply to all students. The experiment tests whether the assumptions necessary for the practical application of this estimation are sufficiently satisfied in this context.

We aimed to predict "success" in the college course for each student. We met with college personnel to decide how to define success, who agreed to define success as a grade of C or better in the initial college-level course associated with the placement decision. We then regressed an indicator for success in the relevant course on various sets of predictors using Probit and linear probability models (LPM). We used the results of the LPMs because we could not code non-linear models into colleges' existing placement software, which had the advantage of producing placement decisions immediately following placement exam completion. The software placed significant constraints on what we could implement because it could largely only incorporate basic, Boolean operators. Nonetheless, the non-linear models yielded similar placement decisions as LPMs around the relevant cut points that colleges chose to determine placements.

For each college, we estimated regressions relating placement test scores, high school GPA[9] and other predictors to "success" in initial college-level classes for a given subject:

$$[1] \qquad [C \text{ or } Better]_i = \alpha_0 + \alpha_1(HS\ GPA_i) + \alpha_2(ACCUPLACER_i) + \boldsymbol{X}'_{ic}\alpha_3 + \epsilon_i.$$

We added additional covariates from high school transcripts when such information was available. This information included the number of years that have passed since high school completion and whether the diploma was a standard high school diploma or a GED (diploma status). We also tested the benefit of including additional variables such as SAT scores, ACT scores, high school rank, indicators for high school attended, and scores on the New York Regents exams, when they were available (often these were missing), as well as interaction terms and higher-order terms for variables. When variables were missing, we imputed a value and added indicators for missing. Identical procedures were followed for both English and math. We estimated the models on prior years of historical data excluding the most recent year and then examined fit criteria using data from that most recent year.[10]

Conditional on placement into the college-level course, exam scores explain very little variation in English course outcomes but more variation in math outcomes; including additional measures adds explanatory power.[11] Appendix Tables A.2 and A.3 list the full set of variables used by each college to calculate students' math and English algorithm scores, respectively. Tables A.4 and A.5 show typical examples of our regression results for math and English. Across colleges, explanatory power is much higher for math course grades than for English course grades. Placement scores typically explain less than 1 percent of the variation in passing grades for English. Test scores are better predictors for passing math grades, explaining roughly 10 percent of the variation. This pattern is likely due to the consistency in content between coursework across colleges in math and math tests. ELA college courses may exhibit more heterogeneity across colleges. Prior research

[9] High school GPA is missing for 37% of students in the experimental sample. This reflects administrative gaps in transcript collection (e.g., older or transfer students, GED completers, or students from schools without automated data pipelines).

[10] The focus of this analysis was the overall predictive power of the model. As such, we calculated the Akaike Information Criterion (AIC) statistics for each model (varying variables in $\boldsymbol{X}_{ic}$). The AIC is a penalized-model-fit criterion that combines a model's log-likelihood with the number of parameters included in a model (Akaike, 1998; Burnham and Anderson, 2002; Hastie et al., 2009; Mazerolle, 2004). Under certain conditions, choosing model specifications according to the AIC is asymptotically equivalent to leave-one-out-cross-validation (Stone, 1977). In practice, we did not have many variables to select from and higher-order and interaction terms had little effect on prediction criteria (and additional complexity was difficult to implement).

[11] The unconditional explained variation differs from what we show in the appendix table. For context, pass rates average around 55% in math and 61% in English.

has also shown that math exams tend to create stronger incentives for test prep (Riehl, 2019; Riehl and Welch, 2022). Adding high school grades typically explains an additional 10 percent of the variation in both subjects. We find that indicators for which high school a student attended, which could reflect different grading standards, add little predictive value. Overall, combining multiple measures with predictive analytics is no panacea for predicting future grades, but it does improve the validity of the placement instrument relative to test scores alone.

Setting Cut Probabilities

After we selected the final models, we used the coefficients from the regression to simulate placement rates for each college using their historical data. Consider the following simplified example where a placement test score (R) and high school GPA (G) are used to predict success in college-level math (Y), defined as earning a grade of C or better. The regression coefficients combined with data on R and G can then predict the probability of earning a C or better in college-level math for incoming students ($\widehat{\mathrm{Y}}$). A set of decision rules must then be determined based on these predicted probabilities. A hypothetical decision rule would be:

$$\text{Placement}_i = \begin{cases} \text{College Level if } \widehat{\mathrm{Y}}_i \geq 0.6 \\ \text{Remedial if } \widehat{\mathrm{Y}}_i < 0.6 \end{cases}$$

For each college, we generated spreadsheets projecting the share of students that would place into college-level coursework at any given cut-point as well as the share of those students we would anticipate earning a C or better. These spreadsheets were provided to colleges so that faculty in the pertinent departments could set cut-points for students entering their programs.

Table A.6 illustrates one such spreadsheet, with separate panels for math and English.[12] Each row reports, for a candidate cut point (a minimum predicted probability of success), the share of students who would place into the college-level course and the share expected to earn a C or better, alongside the status-quo rates. The expected pass rate exceeds the cut point because the cut point is a lower bound: every student placed has at least that probability of passing.

Faculty set cut points that held college-level pass rates at their historical levels. Because the algorithm predicts success more accurately, this generally raised the share of students placed into college-level coursework. In the English panel of Table A.6, a 45

[12] In practice, we showed results from many different cut points.

percent cut point holds the pass rate at its status-quo 60 percent while raising the college-level placement rate from 40 to 75 percent.

Installation of New Placement Method in College Systems

We developed two procedures to implement the algorithms while maintaining the timing of placement decisions. At colleges running our algorithm through the computerized ACCUPLACER-test platform, we programmed custom rules into the ACCUPLACER platform for students selected to be part of the treatment group.[13] These rules created the weights on various student characteristics that, when combined with the colleges' thresholds for placement, produced a placement recommendation for a student.

Other colleges ran their placement through a custom server built for the study. Student information was sent to servers to generate the probability of success and the corresponding placement, which was returned to the college.

3. Experimental Design, Data, Empirical Strategy

We implement a natural field experiment across the participating colleges (Levitt and List 2009; List, 2011). The sample frame consisted of entering cohorts (fall and spring) enrolling at each college who were required to take the placement exams from 2016 until 2018.[14] Random assignment was at the student level and stratified by college. We integrated the assignment procedure into each college's placement platform as described above, such that, upon taking their placement exams, students were randomly assigned to be placed using either the test-score placement system or the algorithmic system. Students and their instructors were blinded to their treatment assignment. This makes the study a natural field experiment (Harrison and List, 2004). If a student took both the English and math placement exams, they were either assigned to the test-score placement system for both subjects or the algorithmic placement system for both subjects. Some students only took a placement exam in one subject. After taking placement exams, students were notified of their placements either by an administrator or through an online portal, depending on the college.

[13] As mentioned above, this process placed constraints on the algorithm's complexity. Interaction terms and non-linear models, for instance, were difficult to implement within the ACCUPLACER system.

[14] Colleges preferred to use alternative placement processes for English as a Second Language speakers, and students with high SAT scores or 4.0 GPA were sometimes exempt from placement exams. Note that, as these are non-selective colleges, few students take the SAT. We report exemption rates in Table 1.

Students overwhelmingly followed their assigned placements. Institutional records indicate that 94 percent of students who enrolled in a math course and 93 percent of those in English took a course at the level indicated by their initial placement. The small share of deviations largely reflects students who retested and received updated placement determinations rather than discretionary overrides. For this reason, our analysis relies on students' initial assigned placement.

The experimental design resulted in a well-powered study, given the constraints. We interviewed faculty and staff to document any perceived changes they saw in the composition of classrooms and any responses to these changes. As we describe below, faculty did not perceive changes to their classroom compositions and so did not make changes to the curriculum or teaching. Given that prior evidence suggests that tracking can allow instructors to target instruction more effectively (Card and Giuliano, 2016 and Duflo et al., 2011), our results might present a lower bound on effectiveness if instructors were to change their behaviors in response to more significant changes in their classroom compositions.

Data

Data came from three sources: placement test records, administrative data from each college, and qualitative data on implementation and quantitative data on costs were collected from faculty, counselors, and staff using interviews and focus groups. Student-level placement test records include indicators for each student's placement level in math and English, as well as the information that would be needed to determine students' placements regardless of treatment status. Therefore, we observe the algorithmic *and* the test-score placement results for every student in our sample, irrespective of whether they were assigned to the treatment or control group. Placement test records from each college contained high school grade point averages (when available) and scores on individual placement tests.

Additional variables included in placement test records varied by each college's placement algorithm. Examples of additional variables incorporated for certain colleges include the number of years between high school completion and college enrollment, type of diploma (high school diploma vs. GED), SAT scores, and New York State Regents Exam scores. In addition to placement test records, college administrative data included demographic information, such as gender, race, age, financial aid status, and transcript data that provided course levels, credits attempted and earned, and course grades.

Table A.7 shows sample baseline characteristics for students who participated in the study at each college and overall. Our sample consists of 12,526 first-year students across the seven colleges. There is some variation in demographic characteristics. For instance, Colleges 1, 2, and 3 serve more white students compared to Colleges 5 and 7, which enroll

a higher share of Hispanic students. Using Pell Grant receipt as a proxy for income, average family income for study participants also varies across colleges; Pell Grant receipt ranges from 32 percent to 57 percent of students. Comparing these characteristics to Appendix Table A.1 shows that the study sample characteristics match the overall characteristics of students each college serves.

Outcomes

We study the effects of assignment to the placement algorithm on several primary outcomes, by subject. First, we examine how placements change as a result of the algorithm: what share of treated students had their placement change relative to the status quo, and of these, what share had their placement change from a remedial-course assignment to a college-level assignment, and what share had their placement change from a college-level course assignment to a remedial assignment. Second, we show treatment effects on enrollment and pass rates for math and English separately. Lastly, we study college and remedial credits attempted and completed. We show these results in the short run—the first term after placement—as well the longer run for the subsample of students we observe for more than two years.

Empirical Strategy

We use an intent-to-treat analysis to examine the impacts of using algorithmic placement versus the test-score placement system. We estimate the following model:

$$[2] \qquad Y_{ic} = \beta_0 + \beta_1 Treatment_{ic} + \boldsymbol{X}'_{ic}\beta_2 + \gamma_c + \varepsilon_{ic},$$

where Y_{ic} are academic outcomes for student i in college c, such as placement into a college-level course and passing a college-level course; $Treatment_{ic}$ indicates whether the individual was randomly assigned to be placed using the algorithmic placement system or the test-score placement system; γ_c are college (strata) fixed-effects; $\boldsymbol{X}_{ic}$ is a vector of baseline covariates (gender, race, age, financial aid status), including students' math and English algorithm scores, which are baseline measures of academic preparedness, and ε_{ic} is the error term. The coefficient of interest is β_1, which is the effect of assignment to the placement algorithm on outcomes discussed above. We estimate Huber-White-Heteroskedasticity robust standard errors following the experimental design (Abadie et al., 2020; Huber, 1967; White, 1980).

As not everyone takes a placement exam in both subjects, we estimate these regressions for those who took any placement exam (which cannot be affected by algorithmic assignment), and therefore are assigned to placement by the algorithm for one *or* two courses. We also estimate these regressions for those who took placement exams

in both subjects and therefore can be assigned to placement by the algorithm for two subjects.

Subgroup Analyses

We also study several differential effects the placement algorithm has on the composition of students placed into remedial and college-level courses. We estimate equation [2] above for each subgroup and test the significance of the interaction terms, shown below.

$$[3] \qquad Y_{ic} = \beta_{0k} + \beta_{1k} Treatment_{ic} + \beta_{2k} Treatment_{ic} \times Subgroup_k + \gamma_c + \boldsymbol{X}'_{ic}\beta_{3k} + \zeta_{ic}.$$

The outcomes, Y_{ic}, are placement in college-level math, placement in college-level English, and credit accumulation. For each k subgroup of interest, we restrict the sample to the reference group and the subgroup. Therefore, the coefficient β_{1k} shows the effect for the reference group (listed below), and the coefficient of particular interest is the significance and magnitude of β_{2k}, which indicates whether the difference between groups of students is widening or narrowing because of algorithmic placement. The subgroups of interest are Black students and Hispanic students compared to white students; female students compared to male students; and Pell recipients compared to non-Pell recipients. This process yields many tests, which increases the likelihood of type-I errors (List et al., 2019; Romano and Wolf, 2005; Scheffé, 1959; Tukey, 1953). To control the family-wise error rate, we use a simple step-down procedure formulated by Holm (1979).

Treatment-Control Baseline Balance

Valid randomization should ensure that, in expectation, students assigned to the treatment group are similar to those assigned to the control group. Table A.8 provides evidence that random assignment was successfully implemented. Participants' demographic and academic characteristics are balanced across treatment and control groups. Students' ACCUPLACER exam scores also are similar across both groups. Overall, the magnitudes of differences between treatment and control groups are small and only two are significant at the 5 percent level, which is unsurprising given the more than 20 variables tested. Though not shown, this balance also holds for the subgroup of students who took both the English and math placement exams as well.

4. Results

Descriptive Changes in Placements

We begin with a descriptive summary of placement changes to show the various ways the algorithm changed students' placements relative to the test-score placement system. As stated above, it is not obvious how the algorithm will change net placement rates.

Table 1 summarizes these changes for students placed by the algorithm. Of the more than 6,000 students assigned to the treatment group, 82 percent were tracked in math and 100 percent were tracked in English. Among those students who took a math placement exam, 23 percent experienced a math placement different from what would have been expected under the test-score placement system. Of those with a changed math placement, 67 percent were placed into a higher-level math course than would have been expected under the test-score placement system, and 33 percent placed in a lower-level math course. Of those who took the English placement exams, approximately 55 percent of treatment-group students experienced a change in the level of their English level placement, of which 90 percent placed into a higher-level English course and 10 percent placed into a lower-level course than they would have under the test-score placement system.

Treatment Effects on Placement, Course Taking, and Credits

Algorithmic placement causes increases in placement into college-level courses, enrollment in college-level courses, and total college-level credits earned.

First-term placement, enrollment, and performance

Table 2 summarizes the first-term results. Students assigned to the placement algorithm are 6.6 percentage points more likely to be placed into a college-level math course, 2.6 percentage points more likely to enroll in a college-level math course, and 1.9 percentage points more likely to pass a college-level math course during the first term. All of these results are statistically significant at the 1 percent level. As described above, one explanation for the difference between placement and enrollment into a college-level math course is that students placed into college-level math do not have to complete a college-level math course prior to enrolling in other college-level courses in the first term.

There are positive and substantially larger effects for English placement, enrollment, and completion than for outcomes on math courses. Students who were placed by the algorithm are 32 percentage points more likely to place into a college-level English course, 14 percentage points more likely to enroll in a college-level English course, and 7 percentage points more likely to pass a college-level English course in the first term. All results are significant at the 1 percent level. Again, the difference between placement and enrollment into a college-level English course may occur for the same reason as above for college-level math enrollment.

Remedial Credits and Credit Accumulation

We also find reductions in total remedial credits taken and increases in total college credits earned. These effects are generally larger for students who are placed via the algorithm in both math and English. The first three columns in Table 3 show results for students who took any placement exam while columns four through six show results for students who took a placement exam in both subjects (and so are tracked in both courses).

Table 3 shows algorithmic assignment reduces remedial credits attempted by 1.1 credits relative to a mean of 3.5 credits—a 31 percent reduction. The effect is almost the same for those tracked in both English and math, but, relative to a mean of 4.1 credits, represents a 26 percent reduction. As we discuss further below in our cost analysis, this reduction in remedial credits reflects a potential savings to students.

Table 3 also shows there is an increase in credit accumulation of 0.56 credits for those tracked in at least one subject and 1.3 credits for those tracked in both subjects.[15] At the outset, it was possible students might substitute math and English college credits for other college-level credits and not just for remedial credits. However, Table A.9 shows the net positive effects on credit accumulation are also due to an increase in other (non-math and non-English) college-level credits and not just a substitution between types of credits.[16]

Pass rates

Colleges chose to hold predicted pass rates constant, and we see that the increase in college-level placement does not change realized pass rates relative to the control group. We calculate realized pass rates as credits earned divided by credits attempted. For students who are tracked in at least one subject, the control group passes 64 percent of their college-level credits attempted while the treatment group pass rate is 63 percent. For those students tracked in both subjects, the control group pass rate is 62 percent and the treatment group pass rate is 63 percent. To formally test this, we estimate the intent-to-treat effect of the algorithm on pass rates for math, English, non-math/non-English, and overall credits in the first term and ever. Across all specifications, we find no statistically significant impact on conditional pass rates.

Improving prediction quality shifts colleges toward the frontier on the tradeoff between placement rates and pass rates: a more accurate model supports a higher placement rate at any target pass rate (Kleinberg et al., 2018). Each college held its target pass rate fixed when adopting the algorithm and adjusted only the placement threshold (Section 2). The increases in placement rates we observe therefore reflect a shift toward the frontier and

[15] Table 3 includes all attempted credits across all subjects, not limited to math or English.

[16] Pass rates for the non-math and non-English credits are the same (around 65%) as for math and English credits.

not a tradeoff between access and pass rates.

Another indicator of performance relative to test-score based placement is what happens when the two systems disagree on placement into college-level courses. We observe counterfactual placements for each student. The first row of Table A.10 shows realized pass rates in each subject when the algorithm recommends placement into college level but the test-score placement system does not. The second row of Table A.10 shows pass rates in each subject when the test-score system recommends placement into college level, but the algorithm does not. The first two columns show pass rates in math and English during the first term, and the second two columns show pass rates for students who ever enroll in the college-level math and English course. When the systems disagree, pass rates are 10 percentage points higher for algorithmic placement in math and 12 percentage points higher in English.[17]

Medium-Run Outcomes

The overall pattern of results holds over the medium or longer run as well. Table A.13 shows results for our earliest cohort of students 2.5 years later. It is possible that, after more than two years, students in the control catch up to students in the treatment group. The control group for this cohort has higher total credit accumulation than the overall sample, as expected, but the increase in total credits earned and the decrease in remedial education credits earned are each larger than what is observed in the short run. These patterns suggest that the short-run gains do not immediately reverse, though these longer-run estimates are based on a single cohort and should be interpreted with caution.

We also estimate these longer-run treatment effects separately by subject. As shown in Appendix Table A.14, the benefits of algorithmic placement are concentrated in English courses. The algorithm substantially reduces remedial enrollment and increases both college-level credits attempted and earned in English, while effects in math are smaller and generally not statistically significant. These results are consistent with the earlier evidence that placement-test scores contain little information about English performance but remain somewhat predictive in math. Together, the findings suggest that the efficiency gains from algorithmic placement are driven primarily by improvements in English placement, implying that the returns to replacing test-based placement may be greater for English than for math.

We also examine the completion of gatekeeper courses and the 60 credit requirement for an associate degree. Algorithm-placed students were more likely to both enroll in and successfully complete key college-level gatekeeper courses in math and English (see Table

[17] We further evaluate the predictive performance of the algorithmic placement system with confusion matrices and classification metrics (Tables A.11 and A.12), and ROC curves (Figures A.1 and A.2), but note that these metrics are calculated on *selected* outcomes, and should be interpreted with caution.

A.15). Despite this increase in completion, and the increase in total credits, algorithmic placement does not raise the probability of earning at least 60 college-level credits, which is the credits required for an associate degree, within five terms (Table A.13, columns (4) and (8)).[18] The estimate is precise enough to rule out effects above 2.4 percentage points on a control mean of 0.090. This is a similar result to the credit gains observed for instructor incentives (Brownback and Sadoff, 2020).

5. Equity, Fairness, and the Implications of Selective Labels

Equity

One measure of equity is how algorithmic placement affected differences in placement rates across subgroups. Table 4 shows the subgroup effects on placement outcomes and credit accumulation for each subject and subgroup. Each cell is a separate regression restricted to the subgroup specified in the column header. The control group mean and the observation count for the outcome and subgroup of interest is shown immediately below the standard error.

The effects of the algorithm on placement into college-level math and English are large and positive for all subgroups, except male students in math. Remedial credits earned also decrease for all subgroups (including male students), and college credits increase a statistically significant amount only for female students, and Pell-grant recipients.

College administrators were interested in how the algorithmic placement system affected differences in placement rates across subgroups. Given this interest, we focus on the extent to which algorithmic placement widened or narrowed differences in key outcomes across subgroups. This question implies we are interested in the interaction terms from equation [3], which tests whether there are differential effects for Black and Hispanic students (separately) relative to white students, female students relative to male students, and Pell recipients relative to non-Pell recipients. Including outcomes in placement for math and English and credit accumulation in remedial and college-level courses, there are 16 interaction terms of interest. We use the step-down method from Holm (1979) to (conservatively) control the family-wise error rate at the five percent level.

Four interaction terms remain significant after this adjustment. Placement rates for Black students into college-level English increase relative to white students and placement rates for female students into college-level math relative to male students increase as well. Though placement rates into college-level math and English courses increase for Hispanic

[18] We could not analyze credential attainment directly, but a follow-on paper finds no effects on those outcomes, in line with our results (Kopko and Daniels Sarica, 2026).

students overall, the increase in math is smaller than it is for white students. Lastly, the decrease in remedial credits is significantly larger for Pell recipients than it is for non-Pell recipients. Thus, students from these groups seem to benefit from algorithmic placement, on average, and there is evidence that many of the benefits accrue to students traditionally under-represented in college courses.

One concern is whether algorithmic placement increases placement rates for certain subgroups but reduces pass rates in college courses for those groups. As mentioned, relative to the test-score placement system, college-level pass rates remain constant overall under the new placement system, but algorithmic placement could have differential effects on college-level pass rates across subgroups.

To check this, we independently estimate equation [2] for each subgroup on pass rates and present the results in separate columns of Table A.16. While there seems to be a slight reduction in pass rates for white students in English and male students in math and English, the few (4 out of 28) statistically significant coefficients become non-significant (at the five percent level) after adjusting for multiple testing using the Holm's step-down method. Overall, pass rates in college-level courses are similar across subgroups.

Assessing Bias

While the algorithm does not include any group membership indicator (i.e., there is no disparate treatment), a placement score may nonetheless be biased. We use what is sometimes called disparate impact or unwarranted disparities (Baron et al., 2024a; Baron et al., 2024b). In the context of course placement, this criterion means that a score, here a test score or an algorithm score, is biased if people from two different demographic groups have different expected placements *conditional on their likelihood of passing the college-level course*. Following the notation of Arnold et al. (2021), a score is biased if:

$$\begin{aligned}[4] \qquad \Delta = (E[P_i \mid Y_i^* = 0, R_i = a] - E[P_i \mid Y_i^* = 0, R_i = b])(1-\mu) \\ +(E[P_i \mid Y_i^* = 1, R_i = a] - E[P_i \mid Y_i^* = 1, R_i = b])\mu \neq 0.\end{aligned}$$

P_i is an individual's placement, based on either their test score or algorithm score. Y_i^* indicates whether a student would take and pass the class, if directly placed. R_i indicates a particular demographic group, and μ is the average pass rate across students in either group. The first term on the righthand side of the equation represents the difference in expected placements between groups a and b conditional on students who *would not* pass the course if placed. The second term on the righthand side represents the difference in expected placements between the same groups conditional on students who *would* pass the course. These two differences are weighted by the overall share of students who would

not pass the course and the overall share of students who would pass the course, respectively. A Δ that is different from zero indicates bias. For example, let P_i be the placement decision under ACCUPLACER system used to place students into college-level math, let demographic group a indicate white students, and let demographic group b indicate Black students. If $\Delta > 0$, this implies Black students who are equally qualified for the college-level course are nonetheless less likely to be placed into the college-level course under the ACCUPLACER system than equally qualified white students.

The challenge in estimating Δ is that we only observe Y_i^*—whether a student passes the college-level course if placed—for a selected set of students. This set is selected because placement into the college-level course is conditional on sufficiently high placement scores. The experiment helps solve this selection problem because the algorithm and test-score placement systems place different sets of students into the college-level course, and the experimental design selects a random sample of students from each of these groups.

For instance, imagine the algorithm selects one set of students, and the test-score system selects the complement of this set. If we only saw one of these systems implemented, we would still only observe Y_i^* for those who meet the placement criterion for that system. However, since we implement both systems simultaneously and randomly assign students to each system, we observe a random sample of students placed by either system. In this example, we would see a random sample of all students placed into the college-level course because the two systems select complementary sets of students. While the latter solves the selection problem, for most colleges it is not true: there remains a large share of students whom neither system would place into the college-level course. However, one college in the study comes close to this scenario in English and nearly does so in math.

We use this college to examine bias in both the ACCUPLACER and algorithm placement systems, as well as to understand the selective-labels problem for predicting college-course performance. At this college, 99 percent of all students eligible for placement in English (N = 905) would have been placed into college-level English by at least one of the placement systems, and 93 percent of students eligible for placement in math (N = 1,679) would have been placed into college-level math by at least one of the placement systems.[19]

This setup is illustrated in Appendix Figure A.3. In the treatment group (Algorithm Placement), a representative sample of students who would have been placed by the *Algorithm* and the *Algorithm & Test Score* in college-level course are observed in college-level courses. In the control group (Test Score Placement), a representative sample of students who would have been placed by *Test Score* and the *Algorithm & Test Score* are

[19] Note that this does not imply that either 99% of students in English or 93% of students in math were placed into the respective college-level course either within or across the two systems.

observed in college-level courses. This design allows us to approximate the thought experiment, preventing the analyses being confounded by a large group of students who would never be placed into college-level courses under either mechanism.

We use students' observed placements across the two systems to estimate Δ as follows:

$$[5] \quad \widehat{\Delta} = \left(\frac{\sum_{n_a=1}^{N_a} P_{ia}(1-Y_{ia}^*)}{\sum_{n_a=1}^{N_a}(1-Y_{ia}^*)} - \frac{\sum_{n_b=1}^{N_b} P_{ib}(1-Y_{ib}^*)}{\sum_{n_b=1}^{N_b}(1-Y_{ib}^*)}\right)\left(1 - \frac{1}{N}\sum_{n=1}^{N} Y_i^*\right) + \left(\frac{\sum_{n_a=1}^{N_a} P_{ia}Y_{ia}^*}{\sum_{n_a=1}^{N_a} Y_{ia}^*} - \frac{\sum_{n_b=1}^{N_b} P_{ib}Y_{ib}^*}{\sum_{n_b=1}^{N_b} Y_{ib}^*}\right)\left(\frac{1}{N}\sum_{n=1}^{N} Y_i^*\right).$$

P_{ia}, P_{ib}, Y_{ia}^*, and Y_{ib}^* are placement decisions (by the algorithm or ACCUPLACER systems) and an indicator for passing the course, respectively, conditional on student i belonging to demographic group a or b. Lastly, $\frac{1}{N}\sum_{n=1}^{N} Y_i^*$ is an estimate of the pass rate if all students were placed into the college-level course.

While our design helps address the selective labels problem, we still do not observe Y_i^* for all students.[20] We impute these missing values for Y_{ia}^* and Y_{ib}^* to either minimize or maximize the difference, $\widehat{\Delta}$, for each system.[21] We then bootstrap these estimates 1,000 times to obtain standard errors of Δ. We estimate $\widehat{\Delta}$ between different demographic groups for the ACCUPLACER and the algorithm systems in math and English. Table 5 shows the resulting estimates and standard errors.

For the ACCUPLACER math placement system, we find evidence of bias across multiple demographic groups. Focusing on the lower bound of each pair of estimates, Black students who are equally likely to succeed in college math as white students have a 15 percentage point lower probability of being placed into a college-level course (80 percent of the raw difference in placement rates between the two groups).[22] Hispanic

[20] Y_i^* is missing 1% in English and 7% in math.

[21] We could tighten these bounds further by assuming pass rates are weakly higher for students with higher test scores or higher algorithm scores. Let pt, pa be the placement decisions by the test score system and algorithm system, respectively, which can take on a value of either 1 for college level or 0 for remedial. Dropping the conditioning on race, $E[Y^*] = E[Y_i^*|p_t = 0, p_a = 0]E[p_t = 0, p_a = 0] + E[Y_i^*|p_t = 1, p_a = 0]E[p_t = 1, p_a = 0] + E[Y_i^*|p_t = 0, p_a = 1]E[p_t = 0, p_a = 1] + E[Y_i^*|p_t = 1, p_a = 1]E[p_t = 1, p_a = 1]$. We do not observe the first term—pass rates for students who had low test *and* low algorithm scores—however, it may be reasonable to assume it is not a higher value than any of the other terms.

[22] For context, Black and Hispanic students have a 22 and 13 percentage point lower probability of being placed into an English college-level course by the ACCUPLACER system, respectively. In comparison, female students have a 2-percentage point higher probability than male students. In math, Black, Hispanic, and Female students have an 18, 16, and 13 percentage point lower probability, respectively, of being placed into college-level by the ACCUPLACER system. Differences in algorithm scores are much smaller across groups: Black and Hispanic students have the same probability as white students to be placed in an English college-level course, and an 8 and 2 percentage point lower probability of being placed into a math college-level course (the last is not statistically significant). Female students have a 0.4

students have a 14 percentage point lower probability of placement into college level math compared to white students who are equally likely to pass (87 percent of the difference in placement rates between the two groups). Similarly, female students have a 13-percentage point lower probability of placement into college-level math than male students (96 percent of the difference in placement rates between the two groups). All these results are statistically significant at the 1 percent level.

The math *algorithm* exhibits smaller bias. For Black students compared to white students, the two estimates of $\widehat{\Delta}$ are 6.0 and 6.8 percentage points in favor of white students (both significant at the 1 percent level). For Hispanic students compared to white students, the estimates are 1.4 and 2.4 percentage points, neither of which is statistically significant. Similarly, for female versus male students, the estimates of $\widehat{\Delta}$ are 2.0 and 1.9 percentage points, neither of which is statistically significant.

For English ACCUPLACER placement, the results are more variable. For Hispanic and Black students compared to white students, $\widehat{\Delta}$ is 13 and 21 percentage points, respectively, which are substantially lower probabilities of being placed into a college-level course than white students conditioning on their outcomes in the course (96 percent and 99 percent of the raw differences in placement rates, respectively). Female students, however, have a 2.7 percentage-points *higher* (not statistically significant) probability of being placed into a college-level English course than male students.

The algorithm for English exhibits similar directional bias for female versus male students: $\widehat{\Delta}$ is 0.5 percentage points in favor of female students (not statistically significant). For Hispanic students compared to white students, there is no difference in the likelihood of being placed into a college-level English course. For Black students compared to white students, $\widehat{\Delta}$ is 0.5 percentage points in favor of white students (not statistically significant).

Though the results above assess a measure of discrimination across the placement systems, the extent to which they exhibit any bias is still within the context of each college's particular placement system and its associated rules and practices. For instance, colleges may change the set of tests taken by providing exemptions from taking the placement tests due to prior credits earned in high school, high SAT scores, or high New York Regents test scores. Colleges may offer formal or informal test preparation guidance to students as well. While these policies tend to be similar across the colleges in our sample, their similarity (or not) to colleges outside the context of our study could limit the external validity of our results.

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percentage point higher probability (non-significant) than male students on the likelihood of being placed into an English college-level course, and 3 percentage point lower probability of being placed into a math college-level course.

The Implications of Selective Labels

We also use this college to examine how the selective labels problem affects our ability to predict success in the college-level courses. To do so, we regress an indicator for passing the college-level course on baseline covariates. The college's placements across both systems, combined with random assignment, allow us to add an important interaction term: an indicator for whether a student would have been placed into college-level by the test-score system. The latter group of students, those placed into the college-level course, is the group typically used for estimating a model predicting course outcomes (cf. Scott-Clayton, 2012 and Scott-Clayton et al., 2014). Without the experiment, these students would be the only ones for whom we observe the relevant outcome because they were placed into the college level course.

Specifically, we estimate the following model that compares coefficients from the "selected model," which conditions on placement into the college level by test-score system, with those who are not placed into college level:

$$[6] \qquad Y_i = \gamma_0 + \boldsymbol{X}_i'\gamma_1 + selected_i\gamma_2 + selected_i * \boldsymbol{X}_i'\theta + \omega_i,$$

Y_i is an indicator for passing the college-level course, $\boldsymbol{X}_i$ is a vector of baseline covariates, such as high school GPA and diploma type, and $selected_i$ is an indicator for whether the test-score placement system would have placed student i into the college-level course. The coefficients of interest are represented by the vector θ, which show whether predictors are weighted differently among those placed into college level by the test-score system versus those not placed into college level. One test of whether the selective labels problem "matters" is reflected in the significance of these interaction terms. Significant differences could imply that the selected model, which conditions on placement into the college level, will make different predictions than an unselected model that does not condition on placement. Table A.17 shows the resulting coefficients and standard errors from estimating [6] for predicting success in college-level English and math.

Several interaction terms are statistically significant. For college English, ACCUPLACER scores and high school GPA have smaller coefficients in the selected model. We fit a Seemingly Unrelated Regression model to conduct a joint test of whether the coefficients estimated using the unselected sample—not conditioned on placement—are equal to the coefficients estimated using the selected sample. We reject this hypothesis at the 1 percent level. For college math, GED status is significantly larger in the selected sample. A joint test for the equality of coefficient estimates across the two models for math rejects this hypothesis at the 10 percent level.

We also use these model estimates to generate predictions of success in college English and math for each student. For English, the predicted probabilities generated from the

unselected model are only weakly correlated with those generated from the selected model; the correlation is 0.22. The correlation of the predicted probabilities for math are much higher, however: 0.94.

What implications do these differences have for prediction errors? To answer this question, we randomly sort the data and partition it into five "folds." We then fit each model on four of the folds and compute the Mean Squared Error (MSE) on the remaining fold (the "hold-out" or "test" sample). We repeat this process by holding out each of the five folds and then computing the MSE each time on the test sample. We average the MSE across the five folds to compute the out-of-sample Mean Squared Error (OOSMSE) for the selected and unselected models in each subject.

In English, the OOSMSE is nearly twice as high for the selected model (the model estimated on the sample that conditions on placement into the college level course via the test-score system) as the unselected model. The OOSMSEs for each model are 0.39 and 0.23, respectively. In math however, the OOSMSEs are nearly identical across the two models: 0.23 for the selected model versus 0.22 for the unselected model. Overall, the evidence suggests that the selective labels problem is more consequential for predicting college English outcomes than predicting college math outcomes.

Placing Students According to Predicted Benefits

Often, algorithms place students based on their predicted outcomes. The experiment allows us to provide suggestive evidence on an alternative algorithm based on predicted *treatment effects*. How would placements differ if we placed students based on whether they are predicted to benefit from remedial coursework over college-level coursework? The experiment creates a subset of students whose placement was effectively randomized; because we know what someone's placement would be under each system, irrespective of their realized assignment, we can subset to the students whose placement would be the college-level course under one system and the remedial course under the other system. 2,913 students had their placement randomized in math and 6,829 students had their placement randomized in English.

We use this subset of students and a generalized random forest approach (Athey et al., 2019) to estimate the conditional average treatment effects (CATE) on total credit accumulation, and predict which students benefit most from remedial course placement (or, symmetrically, we could use college-level placement) in each subject. We then simulate an algorithm that places students according to which course they benefit from most. Given the assumptions to estimate heterogeneous treatments and extrapolate them to all students, we consider the following evidence suggestive.

Tables A.18 and A.19 suggest that few students benefit from remedial coursework. If students were placed according to an algorithm that assigns students according to which

course they benefit from most, even more students would be placed into college-level coursework: 90 percent would be placed into college-level math and 94 percent would be placed into college-level English. 63 percent of students would have their placements changed in math and 21 percent in English relative to an algorithm that places students based on their likelihood of passing the college-level course. That so few students benefit from remedial coursework also suggests that policies that aim to reduce the opportunity cost of remedial coursework by allowing students to take that course at the same time as the college-level course may nonetheless be harming many students (Ryu et al., 2022). Instead, it would be more constructive to redesign remedial coursework to be more effective (Schudde and Keisler, 2019).

6. Cost Analysis

In this section, we present the cost-effectiveness analysis for the algorithmic placement system and the test-score placement systems for six colleges using the ingredients method (Levin et al., 2017). We could not collect complete cost data at one college.[23] The cost estimates reflect the annual expected cost during the first five years of implementing and operating the new placement system at colleges of similar size and organization as the six sample colleges. We summarize the results here and provide more detail below.

Algorithmic placement reduced remediation costs for students: students took fewer remedial credits, saving $220 per student in remedial tuition, and voluntarily attempted more college-level credits, raising their tuition payments by about $260 per student. Relative to the test-score placement system, implementation and operation costs were larger for colleges, $140 per student; operating costs, however, are $40 per student over the status quo. On the margins the placement systems directly target—remedial and college-level math and English credits—total costs are lower for the algorithmic placement system. Once the induced college-level credits in other subjects are included, total educational expenditure rises by about $240 per student, reflecting greater investment in credit-bearing coursework rather than spending on remediation. Following Hendren and Sprung-Keyser (2020), we conduct a comparative welfare analysis by estimating the Marginal Value of Public Funds (MVPF): students' willingness to pay for algorithmic placement divided by the costs to the government. We estimate an MVPF of approximately 1.1, a conservative lower bound.

Costs could be reduced substantially if data to estimate the algorithms did not have to be hand entered and if data collection were centralized into a single system. We detail

[23] What we could collect does not suggest this seventh college had costs significantly different from the others, but personnel changes prevented us from collecting all the necessary data.

the calculations underpinning these findings below.

Defining Costs and Cost Data

To better understand the details of our cost-effectiveness analysis, we start by defining several terms. First, *fixed costs* are those costs that do not vary with college enrollment. *Direct costs* are the costs of implementing and operating the placement system. *Implementation costs* include one-time costs incurred to develop and test the placement method (e.g., evaluator time) and the operating costs to keep it fully functional. *Operating costs* refer to running a placement system after the initial method has been developed and tested (i.e., personnel, facilities, administering placement test, etc.). *Indirect costs* are associated with the price and quantity of credits attempted by the students. The *total costs* are the sum of the *indirect* and *direct costs*. *Student costs* include only the cost of the credits attempted and not the *direct costs*, as students do not pay for the additional costs of implementing the algorithmic placement system. In contrast, *college costs* include *direct costs* of implementing the alternative system and any costs from course offerings (e.g., changes in the number of remedial courses offered). Finally, *cost-efficiency*, in our context, compares the costs of the algorithmic placement system to the test-score placement system (Levin et al., 2017).

We collected data on ingredients from two primary sources. One source for this information was from direct interviews with faculty and staff who implemented the new testing protocols. The second source for input prices and overhead costs was from secondary sources, such as the Integrated Postsecondary Education Data System (IPEDS), described below.

Sources of Costs in the Placement Systems

Understanding the different cost components of the placement systems helps to distinguish fixed costs from operating costs. The initial investment to implement the algorithm has three components. First, data on students' characteristics (including high school transcripts), placements based on test results, and subsequent college outcomes must be collected. In some colleges, these data are already available, but other colleges required more extensive data collection. Second, data must be analyzed to estimate the new placement algorithm. Third, resources must be allocated to create and implement the new system within the college, which includes training personnel. After the initial investment, implementation requires collecting data from entering students and personnel to assign students to either remedial or college-level courses. For the algorithm, one driver of costs was data entry. Data entry costs were lower if the college had all high school information

pre-loaded into their databases. In contrast, data entry costs were higher if each student's information had to be entered into the computing system individually.

For both placement systems there are costs for administering placement tests. Also, for both systems, future resources may be required as students progress into college-level courses after completing remedial coursework. If more students progress into college-level courses, colleges may have to shift resources toward college courses and away from remedial courses in conjunction with any changes in revenue per student.

College faculty, counselors, and administrators did not indicate significant resource changes with respect to instruction. Potentially, the new placement system may change assignments such that more students are now in college-level classes, which would require more college-level faculty and more sections of college-level courses. However, colleges indicated that faculty could be reassigned from teaching remedial classes to teaching college-level classes, and few changes in class size were anticipated even given the changes in placement rates.

Along with the direct implementation and operational costs, there were also indirect costs associated to the different total number of credits attempted by students under the algorithmic placement system. To compute the indirect costs, we used IPEDS information on the six colleges considered in this analysis. The overall cost per college-level credit and remedial credit was approximately $520 (Barnett et al., 2020).

Cost Estimates

Indirect costs: Table A.20 shows the college-level and remedial credits earned and attempted. Using our estimates of costs per credit, the indirect costs for the test-score placement system are $5,420 per student compared to $5,050 per student for the algorithmic placement system. The lower costs of the latter stem from the net decrease of 0.72 in total credits attempted. Thus, the implementation of the algorithmic placement system results in an indirect cost reduction of $370 per student.

Student costs: Students do not pay all the costs associated with each credit attempted. The relevant costs for students are tuition and fees paid for these credits. Using IPEDS data for the six colleges, the cohort-weighted average for tuition and fees is 39 percent of total expenditures per credit (Barnett et al., 2020). Therefore, of the $520 cost per credit, students pay $200 and government funding covers the remaining $320. Consequently, as shown in Table A.21, students attempted fewer credits in total with the algorithmic placement system relative to the test-score placement system and therefore saved $140.

Direct costs: Table A.22 shows the direct costs to implement and operate the algorithmic placement system and the test-score placement system for five years (amortized over cohorts). For a typical college cohort in the sample of 5,808 students, the cost of implementing the algorithmic placement system is $958,810. The cost of the test-

score placement system is $174,240. These estimates imply an incremental cost per student of $140 for algorithmic placement. The remaining two columns show upper and lower bounds for this cost per student, which ranges from $70 to $360. This variation is driven by substantial fixed costs, so colleges with larger enrollments show much smaller per student costs. One implication of these findings is that costs could be reduced substantially with more efficient, centralized data collection. Minimizing hand data entry and centralizing high school student information into a single data system would help automate the algorithm's estimation and reduce costs.

Total costs: We summarize the total costs, direct and indirect for both students and colleges, for each placement system in Table A.23. Table A.23 covers remedial and college-level math and English credits, the margins the placement systems directly target. On these margins, the total cost per student is $230 less for the algorithmic placement system compared to the test-score placement system: the lower indirect costs due to fewer credits attempted more than offset the higher direct cost (see Table A.22). Including the induced college-level credits in other subjects, total expenditure per student is about $240 higher under the algorithmic placement system.

On these margins, the lower total costs of the algorithmic placement system suggest it is cost-effective from a social perspective relative to the test-score placement system: algorithmic placement is more effective regarding the number of college-level credits earned and its total cost is lower. As shown in Table A.23, the cost per college credit earned is $100 less for the algorithmic placement.

Finally, cost effectiveness from the colleges' perspective is harder to establish. On the one hand, colleges must incur the higher costs to implement and operate the new placement method (as shown in Table A.22). On the other hand, we do not incorporate potential increases in net revenues from the additional coursework. These revenue changes will depend on the characteristics of each institution (e.g., enrollment numbers, funding strategy, etc.), which makes it more difficult to determine these changes relative to the status quo. However, as the algorithmic placement method's total cost is lower and leads to greater credit accumulation, we believe this system is likely cost-effective from each colleges' perspective relative to the test-score placement system as well.

Marginal Value of Public Funds: We compute the MVPF following Hendren and Sprung-Keyser (2020) as students' willingness to pay (WTP) divided by the net cost to the government. In the numerator, WTP equals the direct value of relaxing the placement constraint imposed by the test-score system. We assume this equals the reduction in remedial tuition: $220 per student. Students who take additional college-level credits under algorithmic placement do so voluntarily, so by revealed preference those credits weakly raise utility net of tuition and do not reduce WTP. In the denominator, the government's fiscal exposure is the direct cost of implementing and operating the

algorithm, plus the subsidies the government pays on all induced college-level credits—in math and English and in other subjects—less the subsidies saved on the reduced remedial enrollment. Induced credits enter the denominator regardless of how students value them, because the subsidies are real outlays. We estimate a net cost to the government of approximately $200 per student (Tables A.9 and A.23). The implied MVPF is $220/200 \approx 1.1$. Arguably, this is a lower bound. The numerator omits any surplus from their time saved from not taking an unneeded course.[24]

Lastly, colleges could also save money by not purchasing the ACCUPLACER exams, and administrators asked whether students could be placed via the algorithm as accurately without using these test scores. We examined the extent to which the algorithm would place students differently if test scores were not used for prediction. We find that placement rates would change substantially for math courses, by 18 percent, however, for English courses, only 5 percent to 8 percent of placements would change. This finding is in line with the increased predictive value we find for math test scores over English test scores.

7. Conclusion

Our findings indicate that algorithmic placement, which incorporates multiple measures to predict college readiness, significantly impacts how colleges track students into either college-level or remedial courses. First, algorithmic placement allows colleges to choose cut points that explicitly target predicted placement rates and pass rates. Second, the algorithm leads to changes in the placement of students. Across the seven study colleges, more students were placed into college-level math and English courses without reductions in pass rates in either course. There were particularly large increases in college-level placements in English courses. By several measures, the algorithm is more accurate, more equitable, and less discriminatory than the widely used test-score placement system.

While the algorithm's predictive validity is greater than placement scores alone, the algorithms we developed could be improved. Most notably, our model was constrained by implementation in several ways. To produce rapid placement decisions, we had to embed our algorithm into existing systems, which restricted our modeling choices. We could not

[24] Table A.23 shows a $570 reduction in the instructional cost of remedial credits and a $200 increase in the instructional cost of induced college-level math and English credits; the induced college-level credits in other subjects (Table A.9) add a further $470. Students pay 39 percent of these costs as tuition and the government pays the remainder (Barnett et al., 2020). On the remedial side, students save $220 in tuition and the government saves $350 in subsidies. On the induced college-level credits, students pay $80 more for math and English and $180 more for other subjects; the government pays $120 and $290 more, respectively. Adding the $140 direct placement cost, the net cost to the government is approximately $200 per student, and the implied MVPF is $220/200 \approx 1.1$.

for instance, implement a non-linear model. Future models could also use richer transcript data. The colleges we worked with could not readily provide course-level high school grades, which could be predictive of future performance as well. More generally, as colleges develop more consistent ways to record incoming student information, the ability to predict future performance should improve.

One question is how our results would differ if all students within a college were placed according to the algorithm. Our interviews with college administrators, department chairs, faculty and counselors at each college documented their impressions of the algorithm's implementation. Generally, there was no perceived change in classroom composition. However, this could change if all students were placed via the algorithm, especially in English courses where placement changes were more significant. Moving from a randomized subset to full-scale adoption can itself change effects through general-equilibrium and behavioral channels that a partial-scale experiment does not capture, a central concern when translating experimental findings into policy at scale (Al-Ubaydli et al., 2017, 2020). Prior research suggests this might result in improved academic outcomes for students (Duflo et al., 2011).

Our results have important implications because the high cost of remedial education falls onto students placed into these courses and indirectly onto taxpayers whose money helps subsidize public postsecondary institutions. As a result, there is both a private and social benefit to ensuring that remedial education is correctly targeted. Colleges recognize this, and some have begun to implement these placement algorithms. Long Beach City College (LBCC) created a placement formula that uses student high school achievement in addition to standardized assessment scores. The formula weights each measure based on how predictive it is of student performance in college courses (Long Beach City College, Office of Institutional Effectiveness, 2013). This paper provides evidence that these placement systems not only affect student outcomes through changes in the placement instrument, but also through colleges' improved ability to target pass rates explicitly. Future research could test more intricate predictive models than we could implement in the current study, and perhaps focus on algorithms that predict treatment effects of each course rather than pass rates.

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TABLES

Table 1. Changes in Placement for Treatment-Group Students

	(1)	(2)	(3)	(4)	(5)
	Took Placement Exam	Same Placement as Test Score System	Placement Changed from Test Score System	Higher Placement than Test Score System	Lower Placement than Test Score System
			Math Placement		
% of sample	81.71%	58.58%	23.13%	15.45%	7.68%
N	5,224	3,745	1,479	988	491
			English Placement		
% of sample	100%	45.21%	54.79%	49.05%	5.74%
N	6,393	2,890	3,503	3,136	367

Notes: Sample is restricted to treatment group students: students who took a placement exam in at least one subject and enrolled at one of the seven study colleges during the study period and were assigned to the treatment group. Colleges exempted a share of students from placement tests based on certain criteria, shown in column (1).

Table 2. Effect on Math and English College Coursework

	(1)	(2)	(3)	(4)	(5)	(6)
	Placed Math 1st-Term	Enrolled Math 1st-Term	Passed Math 1st-Term	Placed English 1st-Term	Enrolled English 1st-Term	Passed English 1st-Term
Treatment	0.066***	0.026***	0.019***	0.322***	0.136***	0.070***
	(0.008)	(0.008)	(0.007)	(0.008)	(0.009)	(0.009)
Control Mean	0.376	0.280	0.155	0.492	0.471	0.292
Observations	9,527	9,527	9,527	10,033	10,033	10,033

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Sample is any student who took a placement exam in at least one subject and enrolled at one of the seven study colleges during the study period. Columns (1)-(3) restrict to students who took the math exam. Columns (4)-(6) restrict to students who took the English exam. All models include fixed effects for college (strata), controls for demographic indicators (race, gender and age, Pell recipient status), GED indicator, and calculated math and English algorithm scores.

Table 3. Effect on College-Course Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
	Remedial Credits Attempted	College Credits Attempted	College Credits Earned	Remedial Credits Attempted	College Credits Attempted	College Credits Earned
Treatment	-1.095***	1.287***	0.562*	-1.068***	1.865***	1.312***
	(0.072)	(0.295)	(0.292)	(0.105)	(0.406)	(0.393)
Control Mean	3.539	26.19	16.80	4.141	24.42	15.21
Sample	All	All	All	Placed in Math and English	Placed in Math and English	Placed in Math and English
Observations	12,526	12,526	12,526	6,930	6,930	6,930

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Columns (1)-(3) use the full sample and columns (4)-(6) restrict the sample to students who were placed in both math and English. All models include fixed effects for college (strata), controls for demographic indicators (race, gender and age, Pell recipient status), GED indicator, and calculated math and English algorithm scores. Credits attempted and credits earned are *total* credits attempted or earned by students, respectively.

Table 4. Subgroup Analysis on College-Course Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	White	Hispanic	Black	Male	Female	Pell	Non-Pell
Placed into College Math	0.107***	0.036**	0.054***	0.013	0.140***	0.056***	0.078***
	(0.013)	(0.015)	(0.019)	(0.012)	(0.012)	(0.012)	(0.011)
Control Mean	0.400	0.418	0.287	0.426	0.336	0.331	0.411
Observations	3,778	2,088	1,792	4,369	4,460	4,426	5,027
Placed into College English	0.297***	0.309***	0.385***	0.318***	0.335***	0.289***	0.351***
	(0.013)	(0.019)	(0.019)	(0.012)	(0.012)	(0.013)	(0.011)
Control Mean	0.540	0.505	0.374	0.493	0.491	0.480	0.500
Observations	4,068	2,065	2,039	4,872	4,520	4,565	5,438
College Credits Earned	0.144	0.767	0.918	-0.060	1.186***	1.284***	-0.079
	(0.484)	(0.659)	(0.605)	(0.432)	(0.438)	(0.456)	(0.375)
Control Mean	21.09	16.69	12.44	17.32	18.10	18.88	15.07
Observations	5,379	2,484	2,467	5,951	5,932	5,728	6,798
Remedial Credits Attempted	-0.686***	-0.791***	-0.745***	-0.557***	-0.831***	-0.829***	-0.517***
	(0.065)	(0.114)	(0.114)	(0.064)	(0.071)	(0.075)	(0.054)
Control Mean	1.690	2.227	2.108	1.687	2.029	2.318	1.299
Observations	5,379	2,484	2,467	5,951	5,932	5,728	6,798

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Each column restricts the sample to the subgroup in the column header. Each cell is from a separate regression. All models include fixed effects for college (strata), controls for demographic indicators (race, gender and age, Pell recipient status), GED indicator, and calculated math and English algorithm values. Credits attempted and earned are total credits attempted and earned by students, respectively.

Table 5. Selective Labels: Delta Estimates

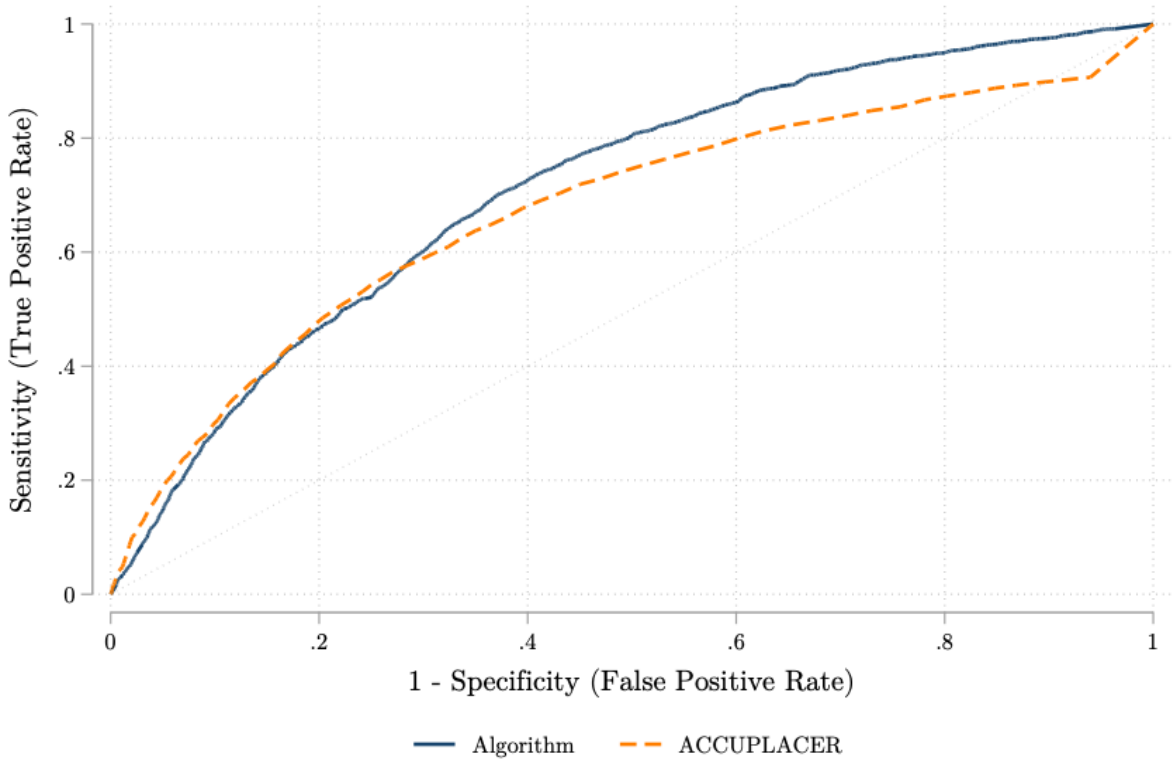
	(1)	(2)	(3)
	White−Black	White−Hispanic	Male−Female
Panel A: Test Score Placement College-level Math			
Δ_1	0.188***	0.154***	0.135***
	(0.033)	(0.030)	(0.025)
Δ_0	0.146***	0.138***	0.128***
	(0.031)	(0.029)	(0.023)
Panel B: Algorithm Placement College-level Math			
Δ_1	0.068***	0.024	0.019
	(0.019)	(0.016)	(0.014)
Δ_0	0.060***	0.014	0.020
	(0.019)	(0.016)	(0.013)
Panel C: Test Score Placement College-level English			
Δ_1	0.213***	0.128***	-0.027
	(0.036)	(0.039)	(0.029)
Δ_0	0.213***	0.128***	-0.027
	(0.036)	(0.039)	(0.029)
Panel D: Algorithm Placement College-level English			
Δ_1	0.004	0.000	-0.004
	(0.004)	(0.000)	(0.003)
Δ_0	0.005	0.000	-0.005
	(0.005)	(0.000)	(0.004)

Notes: Bootstrap standard errors shown in parentheses (reps. = 1,000). *** p<0.01, ** p<0.05, * p<0.10. Each cell is an estimation of Δ from Equation [5] for a placement (specified in the Panel) and demographic pair (specified in the Column). The estimation in each cell is restricted to students in College 5 eligible for placement in Math (Panels A and B) or English (Panels C and D). See Table A.7 for descriptive statistics of College 5. Δ_1 and Δ_0 are calculated by imputing missing values for passing (1) or not (0) the college-level course, respectively.

APPENDIX

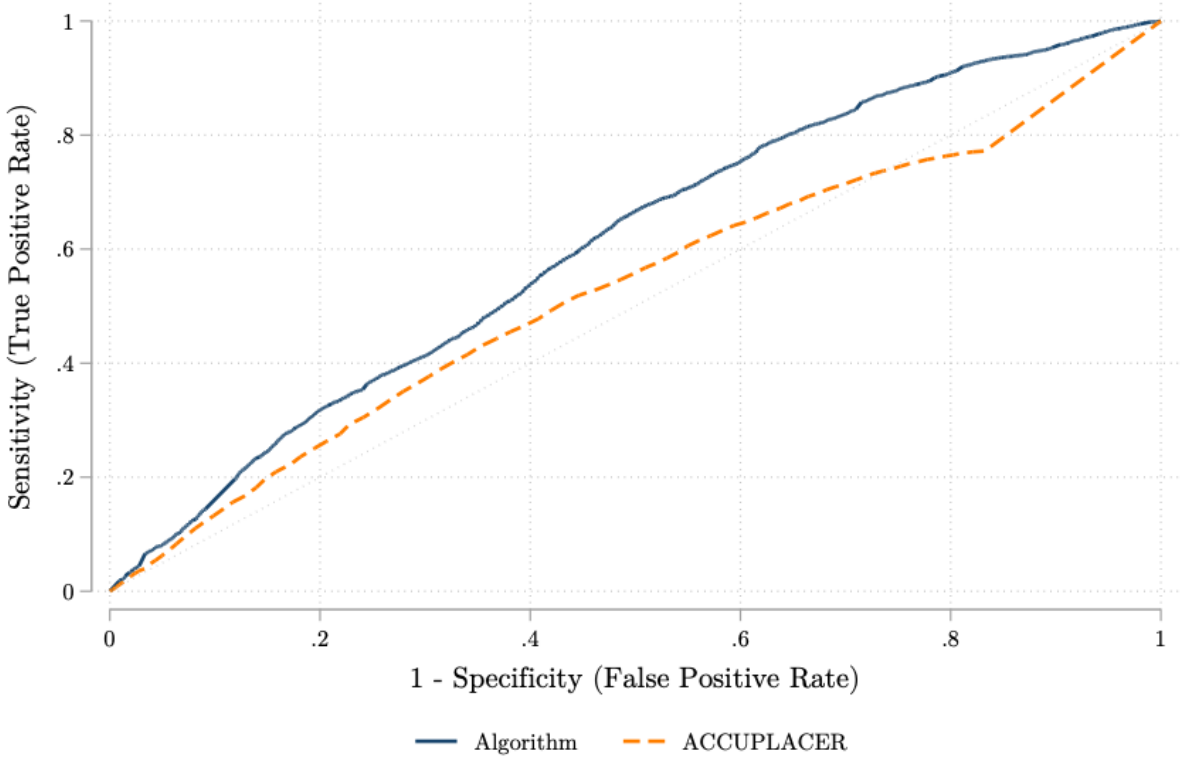
APPENDIX FIGURES

Figure A.1. ROC Curves: Math Placement



Notes: ROC curves estimated among students placed in college-level math courses within each experimental arm. The algorithm curve is estimated on the treatment arm; the ACCUPLACER curve on the control arm. The outcome is whether the student passed the college-level math course. The algorithm score (a predicted probability of passing) serves as the continuous classifier for the treatment arm; the ACCUPLACER College-Level Math score serves as the classifier for the control arm. Algorithm AUC = 0.607, ACCUPLACER AUC = 0.591.

Figure A.2. ROC Curves: English Placement



Notes: ROC curves estimated among students placed in college-level English courses within each experimental arm. The algorithm curve is estimated on the treatment arm; the ACCUPLACER curve on the control arm. The outcome is whether the student passed the college-level English course. The algorithm score (a predicted probability of passing) serves as the continuous classifier for the treatment arm; the ACCUPLACER Reading score serves as the classifier for the control arm. Algorithm AUC = 0.552, ACCUPLACER AUC = 0.526.

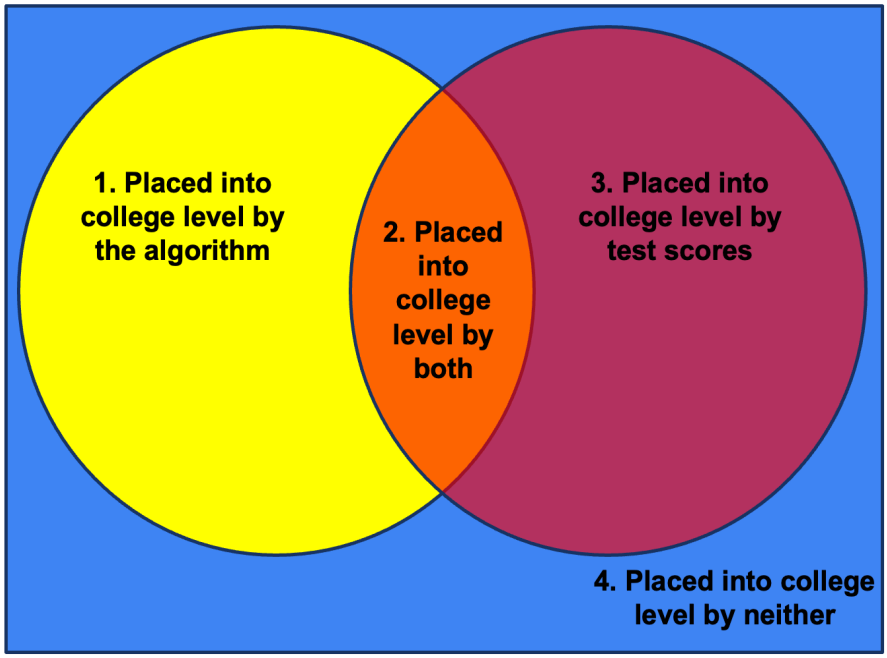
Figure A.3. Fairness Analysis Sample

Key insight:

1. There are four types of people.

2. Random assignment samples from all four of these groups

3. If group 4 is ~0, then all who are remaining would be placed into college level by one system or the other



Notes: The figure illustrates the four groups of students defined by the intersection of algorithmic and test-score placements. The fairness analysis sample consists of students in Groups 1, 2, and 3, those placed into college-level by at least one system, for whom college-level course outcomes are observed. Random assignment ensures that treatment and control students are drawn from all four groups. The analysis exploits the near-absence of Group 4 students to compare outcomes across racial and socioeconomic subgroups without relying on extrapolation.

APPENDIX TABLES

Table A.1. College Characteristics

	Institution						
	Cayuga	Jefferson	Niagara	Onondaga	Rockland	Schenectady	Westchester
GENERAL INFORMATION							
Student Population	7,001	5,513	7,712	23,984	10,098	8,458	22,093
Full-time Faculty	69	80	151	194	122	79	215
Part-time Faculty	170	177	0	480	409	0	2
Student/Faculty Ratio	20	18	16	23	23	23	16
% Receiving Financial Aid	92%	91%	92%	92%	56%	92%	70%
DEMOGRAPHICS							
Race/ethnicity:							
American Indian/Alaska	0%	1%	1%	1%	0%	1%	1%
Asian	1%	2%	1%	3%	5%	7%	4%
Black	5%	7%	11%	12%	18%	14%	21%
Hispanic/Latino	3%	11%	3%	5%	20%	6%	32%
Native Hawaiian or Other	0%	0%	0%	0%	0%	1%	0%
White	85%	73%	80%	49%	39%	67%	33%
Multi-Ethnic	2%	3%	2%	3%	2%	2%	2%
Race/Ethnicity Unknown	3%	3%	1%	27%	15%	2%	5%
Non-Resident Alien	1%	1%	0%	0%	1%	0%	1%
Gender:							
Female	60%	58%	59%	52%	54%	53%	53%
Male	40%	42%	41%	48%	46%	47%	47%
Age:							
Under 18	30%	17%	19%	24%	10%	37%	1%
18-24	44%	52%	60%	55%	63%	40%	69%
25-65	26%	31%	21%	21%	26%	23%	30%
RETENTION/GRADUATION RATES							
Retention							
Full-Time Students	56%	55%	63%	57%	68%	56%	64%
Part-Time Students	28%	30%	47%	34%	56%	50%	53%
Three-Year Graduation Rate	24%	27%	28%	20%	29%	20%	15%
Transfer Out Rate	18%	19%	18%	22%	19%	22%	18%

Notes: This table shows summary statistics for all students enrolled at the seven study colleges from historical data. Data are from the U.S. Department of Education, National Center for Education Statistics, IPEDS, Fall 2015, Institutional Characteristics.

Table A.2. Math Algorithm Components by College

	HS GPA	Years Since HS Graduation	GED Status	Regents Math Score	SAT Math Score	Arithmetic Test Score	Algebra Test Score	College-Level Test Math
College 1	X	X	X			X	X	X
College 2	X	X	X	X	X	X	X	X
College 3	X	X	X			X	X	
College 4								
College 5	X	X				X	X	X
College 6								
College 7	X	X	X				X	

Notes: This table indicates what variables colleges used in their respective math algorithm. Test score variables are from ACCUPLACER placement exams. HS abbreviates high school.

Table A.3. English Algorithm Components by College

	HS GPA	HS Rank	Years Since HS Graduation	GED Status	Reading Score	Sentence Skills Score	Writing Score
College 1	X		X	X	X	X	
College 2	X		X	X	X	X	X
College 3	X		X	X	X		X
College 4	X	X	X	X	X	X	X
College 5	X		X		X	X	X
College 6	X		X	X			
College 7	X		X	X	X		

Notes: This table indicates what variables colleges used in their respective English algorithm. Test score variables are from ACCUPLACER or other placement exams. HS abbreviates high school.

Table A.4. Math Algorithm Models

	Model 1	Model 2	Model 3	Model 4
HS GPA[1]	0.035***		0.028***	0.030***
	(0.002)		(0.003)	(0.002)
Missing GPA[2]	2.822***		2.270***	2.583***
	(0.195)		(0.209)	(0.210)
ACPL Algebra[3]		0.006***	0.004***	0.004***
		(0.001)	(0.001)	(0.001)
ACPL Arithmetic Missing[2]		0.056	0.038	0.065
		(0.040)	(0.041)	(0.042)
ACPL Algebra Missing[2]		0.634***	0.361**	0.335*
		(0.141)	(0.137)	(0.140)
ACPL College-level Math Missing[2]		-0.087	-0.088	-0.084
		(0.055)	(0.051)	(0.051)
Years Since HS Graduation				0.020***
				(0.004)
HS Graduation Year Missing[2]				-0.056
				(0.068)
GED[2]				-0.192**
				(0.071)
Missing Diploma Type[2]				0.121
				(0.100)
Constant	-2.337***	0.038	-2.048***	-2.303***
	(0.192)	(0.122)	(0.217)	(0.213)
N	1,166	1,166	1,166	1,166
R^2	0.125	0.105	0.176	0.207
AIC	1,538.4	1,568.6	1,475.5	1,439.5

[1] 100-point scale

[2] Binary indicator

[3] Test score range 20-120

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. This table shows results from regression of the covariates listed on an indicator for getting a C or better in the college-level math course. Models 1 - 3 include different subsets of covariates, with the full model shown in Model 4.

Table A.5. English Algorithm Models

	Model 1	Model 2	Model 3	Model 4
HS GPA[1]	0.022***		0.022***	0.024***
	(0.001)		(0.001)	(0.001)
Missing GPA[2]	1.774***		1.761***	1.959***
	(0.103)		(0.103)	(0.114)
ACPL Reading[3]		0.001*	0.001*	0.001
		(0.001)	(0.001)	(0.001)
ACPL Sentence Skills[3]		0.000	0.000	0.000
		(0.001)	(0.001)	(0.001)
ACPL Written Exam[4]		0.000	-0.002	-0.001
		(0.002)	(0.002)	(0.002)
ACPL Reading Missing[2]		0.315***	0.332***	0.210**
		(0.073)	(0.074)	(0.077)
ACPL Sentence Skills Missing[2]		-0.027	-0.147*	-0.154*
		(0.077)	(0.074)	(0.074)
ACPL Written Exam Missing[2]		0.021	0.008	0.017
		(0.027)	(0.026)	(0.025)
Years Since HS Graduation				0.009***
				(0.001)
HS Graduation Year Missing[2]				0.041
				(0.087)
GED[2]				-0.190*
				(0.083)
Missing Diploma Type[2]				0.032
				(0.094)
High School Rank Percentile				0.000
				(0.000)
Missing High School Rank[2]				-0.006
				(0.041)
Constant	-1.147***	0.478***	-1.218***	-1.301***
	(0.101)	(0.060)	(0.111)	(0.118)
N	3,786	3,786	3,786	3,786
R^2	0.072	0.006	0.078	0.095
AIC	4,893.2	5,161.4	4,879.8	4,823.8

[1] 100-point scale

[2] Binary indicator

[3] Test score range 20-120

[4] Test score range 1-8

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

This table shows results from regression of the covariates listed on an indicator for getting a C or better in the college-level English course. Models 1 - 3 include different subsets of covariates, with the full model shown in Model 4.

Table A.6. Hypothetical Spreadsheet Provided to Colleges on Placement Projections

Example Community College		
Math Success: C or above		
Minimum Probability of Success	Percent Placed into College Level	Percent Passing College Level
Cohort 3, Status Quo	30%	50%
45%	40%	60%
55%	20%	70%
65%	10%	75%
Eng. Success: C or above		
Minimum Probability of Success	Percent Placed into College Level	Percent Passing College Level
Cohort 3, Status Quo	40%	60%
45%	75%	60%
55%	60%	65%
65%	20%	70%

Notes: This table is a hypothetical version of the information presented to college faculty and administrators to help them choose a threshold for being placed into college-level course in math or English. The placement algorithm outputs a probability of success in the college-level math and/or English course for each student. Colleges then choose what probability is the "minimum probability" acceptable for placement into the college-level course. Several possible minimum probabilities are shown in the leftmost column. The middle column and the rightmost then show the predicted percent of students placed into the college-level course and the predicted pass rate for those students, respectively, associated with the minimum probability shown in the same row. The highlighted row reports the status quo at the college; in this hypothetical example, the status-quo placement and pass rates are 30 and 50 percent for math and 40 and 60 percent for English. Rows below the highlighted row show what would happen at alternative cut points. For math, a 45 percent cut point—meaning a student must have at least a 45 percent predicted probability of earning a C or better in the gatekeeper course to be placed—would place 40 percent of students into college-level math, of whom an anticipated 60 percent would pass. The expected pass rate exceeds the cut point because the cut point is a lower bound: every placed student has at least that probability of passing, and most exceed it. Faculty set cut points that held college-level pass rates at their historical levels; because the algorithm predicts success more accurately, this generally raised placement into college-level coursework. For English, a 45 percent cut point holds the pass rate at its status-quo 60 percent while raising the college-level placement rate from 40 to 75 percent.

Table A.7. Sample Demographics by College

	Overall	College 1	College 2	College 3	College 4	College 5	College 6	College 7
Female	50%	58%	54%	53%	48%	51%	55%	46%
Race								
White	43%	80%	69%	56%	53%	36%	41%	24%
Asian	2%	1%	1%	1%	2%	5%	9%	2%
Black	20%	9%	17%	20%	23%	21%	31%	19%
Hispanic	20%	5%	3%	4%	11%	28%	14%	33%
Native American	1%	1%	1%	1%	2%	0%	1%	1%
Two or more races	3%	1%	3%	4%	6%	3%	3%	3%
Age at entry	20.93	20.83	22.91	22.03	20.24	21.51	23.09	19.92
Pell Grant recip.	43%	52%	47%	49%	42%	32%	57%	42%
Total	12,526	671	1,228	1,817	1,996	1,756	343	4,715

Notes: This table shows the demographic characteristics of study participants in each college. Sample is any student who took a placement exam in at least one subject and enrolled at one of the seven study colleges during the study period.

Table A.8. Baseline Characteristics by Treatment Assignment

	Control Mean	Treatment Mean	Difference (T - C)	P-value	Obs.
Enrollment	0.86	0.85	-0.01	0.29	12,526
Female	0.50	0.50	0.01	0.47	11,883
Race					
White	0.44	0.42	-0.02	0.04	12,526
Asian	0.02	0.02	0.00	0.80	12,526
Black	0.19	0.20	0.02	0.03	12,526
Hispanic	0.19	0.20	0.01	0.16	12,526
Pacific Islander	0.00	0.00	0.00	0.91	12,526
Native American	0.01	0.01	0.00	0.50	12,526
Two or more races	0.04	0.03	0.00	0.24	12,526
Race Missing	0.10	0.10	0.00	0.83	12,526
Age at entry	20.96	20.90	-0.05	0.63	12,526
Pell Grant recip.	0.42	0.43	0.01	0.30	12,526
TAP Grant recip.	0.31	0.31	0.00	0.90	12,526
GED recip.	0.07	0.07	0.00	0.92	12,526
HS GPA (100 scale)	77.96	78.13	0.17	0.32	7,857
HS GPA missing	0.37	0.37	0.00	1.00	12,526
ACCUPLACER Exam score					
Arithmetic	33.35	34.18	0.84	0.16	10,187
Algebra	48.06	47.98	-0.09	0.87	10,187
College-level math	8.25	8.08	-0.17	0.75	3,655
Reading	58.05	58.12	0.07	0.91	12,526
Sentence skills	35.45	34.10	-1.35	0.08	10,709
Written exam	3.87	3.93	0.06	0.32	10,968
Total	6,133	6,393			12,526

Notes: Means and Treatment – Control differences are rounded to the nearest hundredth of a point. Sample is any student who took a placement exam in at least one subject and enrolled at one of the seven study colleges during the study period. The difference column shows estimates from a regression of the baseline characteristic shown on the lefthand side on a treatment indicator and strata fixed effects (indicators for each college). Observation counts vary for exam scores because students do not necessarily take all exams and gender and HS GPA are not available for all students.

Table A.9. Non-English and Non-Math Credits

	(1)	(2)	(3)	(4)
	Non-English/Math Credits Attempted	Non-English/Math Credits Earned	Non-English/Math Credits Attempted	Non-English/Math Credits Earned
Treatment	0.915***	0.424*	1.360***	0.974***
	(0.237)	(0.235)	(0.319)	(0.312)
Control Mean	19.31	12.81	17.64	11.43
Sample	All	All	Placed in Math and English	Placed in Math and English
Observations	12,526	12,526	6,930	6,930

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Columns (1)-(2) are the full sample and columns (3)-(4) restricts the sample to students who were placed in both math and English. All models include fixed effects for college (strata), controls for demographic indicators (race, gender and age, Pell recipient status), GED indicator, and calculated math and English algorithm values. Credits attempted and earned are total credits attempted and earned by students.

Table A.10. Error Rates of the Algorithm v. Test-Score Placement Systems

		(1)	(2)	(3)	(4)
		Math CL-Credits Pass Rate (1st-term)	English CL-Credits Pass Rate (1st-term)	Math CL-Credits Pass Rate (Ever)	English CL-Credits Pass Rate (Ever)
Treatment	Placed in College-Level by Algorithm, Counterfactual was Remedial	0.607	0.626	0.501	0.532
Control	Placed in College-level by Test Scores, Counterfactual was Remedial	0.442	0.496	0.401	0.408

Notes: The first row restricts the sample to students assigned to the treatment group who were placed by the algorithm into a college-level course, but their counterfactual test-score placement would have been the remedial course. The second row restricts the sample to students assigned to the control group who were placed by the test-score system (ACCUPLACER) into a college-level course, but their algorithmic placement would have been the remedial course. Each column shows pass rates by subject for the first term and ever. Pass rates are college-level credits earned divided by college-level credits attempted in the corresponding period and subject.

Table A.11. Confusion Matrices for Algorithmic and Test-Score Placement Systems (Percent)

	(1)	(2)	(3)	(4)
	True Positives	False Positives	False Negatives	True Negatives
Panel A: Math				
Algorithm Placement	21.9	22.0	10.6	45.5
Test-Score Placement	19.8	17.4	11.7	51.1
Panel B: English				
Algorithm Placement	43.8	37.8	5.5	12.9
Test-Score Placement	26.6	19.8	19.9	33.6

Notes: True Positives (TP) = students placed into college-level who passed; False Positives (FP) = over placements (placed college-level but failed); False Negatives (FN) = under placements (placed remedial but would have passed); True Negatives (TN) = correct remedial placements (placed remedial and failed). Each cell reports the percentage of all students eligible for placement in the corresponding subject, with denominator equal to TP + FP + FN + TN.

Table A.12. Classification Performance for Algorithmic and Test-Score Placement Systems (Percent)

	(1)	(2)	(3)	(4)	(5)
	Accuracy	Precision	Recall (Sensitivity)	Specificity	F1
Panel A: Math					
Algorithm Placement	67.4	49.8	67.5	67.4	57.3
Test-Score Placement	70.9	53.2	62.8	74.6	57.6
Panel B: English					
Algorithm Placement	56.7	53.7	88.8	25.4	66.9
Test-Score Placement	60.2	57.3	57.1	62.9	57.2

Notes: Accuracy = (TP + TN)/Total; Precision = TP/(TP + FP); Recall = TP/(TP + FN); Specificity = TN/(TN + FP); F1 = 2 × (Precision × Recall)/(Precision + Recall). Higher recall implies fewer Type II (under placement) errors; higher precision implies fewer Type I (over placement) errors.

Table A.13. Longer-Run Effects: Fall 2016 Cohort

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Remedial Credits Attempted	College Credits Attempted	College Credits Earned	Earned 60+ College Credits	Remedial Credits Attempted	College Credits Attempted	College Credits Earned	Earned 60+ College Credits
Treatment	-1.193***	2.594***	1.717***	0.008	-1.227***	2.659***	2.144***	0.005
	(0.129)	(0.562)	(0.568)	(0.008)	(0.166)	(0.677)	(0.670)	(0.009)
Control Mean	3.916	32.8	21.67	0.090	4.601	30.06	19.00	0.070
Sample	All	All	All	All	Placed in Math and English	Placed in Math and English	Placed in Math and English	Placed in Math and English
Observations	4,685	4,685	4,685	4,685	3,219	3,219	3,219	3,219

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10.
Columns (1)-(4) use the full sample and columns (5)-(8) restrict the sample to students who were placed in both math and English. All models include fixed effects for college (strata), controls for demographic indicators (race, gender and age, Pell recipient status), GED indicator, and calculated math and English algorithm scores. Credits attempted and earned are total credits attempted and earned by students, respectively. Columns (4) and (8) report the effect on an indicator for having earned at least 60 cumulative college-level credits, roughly the credits required for an associate degree; the remaining columns report credits. For column (4), the 95 percent confidence interval is [-0.008, 0.024] and the minimum detectable effect at 80 percent power and a 5 percent size is 0.023 against a control mean of 0.090.

Table A.14. Longer-Run Effects by Subject: Fall 2016 Cohort

	(1)	(2)	(3)	(4)	(5)	(6)
	Math Remedial Courses Attempted	Math College Credits Attempted	Math College Credits Earned	English Remedial Courses Attempted	English College Credits Attempted	English College Credits Earned
Panel A: Placed in Math or English						
Treatment	-0.076***	0.250*	0.154	-0.207***	0.470***	0.280***
	(0.026)	(0.128)	(0.094)	(0.017)	(0.093)	(0.083)
Control Mean	0.720	3.943	2.055	0.383	4.458	2.791
Observations	4,685	4,685	4,685	4,685	4,685	4,685
Panel B: Placed in Math and English						
Treatment	-0.002	0.089	0.135	-0.276***	0.541***	0.374***
	(0.032)	(0.153)	(0.109)	(0.021)	(0.111)	(0.100)
Control Mean	0.817	3.778	1.863	0.486	4.330	2.617
Observations	3,219	3,219	3,219	3,219	3,219	3,219

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Panel A uses the full sample and Panel B restricts the sample to students who were placed in both math and English. All models include fixed effects for college (strata), controls for demographic indicators (race, gender and age, Pell recipient status), GED indicator, and calculated math and English algorithm scores.

Table A.15. Impacts of Algorithmic Placement on Gatekeeper Course Enrollment and Completion

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Enrolled GK Math	Passed GK Math	Enrolled GK English	Passed GK English	Enrolled GK Math	Passed GK Math	Enrolled GK English	Passed GK English
Treatment	0.025***	0.014***	0.048***	0.029***	0.024***	0.022***	0.067***	0.052***
	(0.006)	(0.005)	(0.007)	(0.008)	(0.009)	(0.007)	(0.009)	(0.011)
Control Mean	0.172	0.102	0.652	0.432	0.178	0.101	0.643	0.416
Sample	All	All	All	All	Placed in Math and English	Placed in Math and English	Placed in Math and English	Placed in Math and English
Observations	12,526	12,526	12,526	12,526	6,930	6,930	6,930	6,930

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. All models include fixed effects for each college, controls for demographic indicators including race, gender and age, Pell recipient status, GED indicator, and calculated math and English algorithm values.

Table A.16. Subgroup Analysis on Pass Rates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	White	Hispanic	Black	Male	Female	Pell	Non-Pell
Total Credits' Pass Rate (1st-term)	-0.020*	0.002	-0.017	-0.011	-0.002	-0.012	-0.000
	(0.011)	(0.017)	(0.018)	(0.011)	(0.011)	(0.011)	(0.011)
Control Mean	0.709	0.623	0.534	0.624	0.673	0.643	0.652
Observations	4,686	2,167	2,062	5,134	4,901	5,310	4,725
Total Credits' Pass Rate (Ever)	-0.015	0.010	-0.007	-0.012	0.008	-0.003	-0.001
	(0.010)	(0.014)	(0.014)	(0.009)	(0.009)	(0.009)	(0.010)
Control Mean	0.600	0.472	0.411	0.498	0.555	0.508	0.547
Observations	4,924	2,342	2,221	5,417	5,270	5,703	4,984
Total Math Credits' Pass Rate (Ever)	-0.009	-0.008	-0.039	-0.029*	0.012	-0.012	-0.005
	(0.016)	(0.023)	(0.026)	(0.015)	(0.016)	(0.015)	(0.016)
Control Mean	0.580	0.488	0.445	0.492	0.576	0.514	0.550
Observations	2,940	1,425	1,040	3,200	2,921	3,136	2,985
Total English Credits' Pass Rate (Ever)	-0.036***	0.001	-0.005	-0.028**	-0.001	-0.010	-0.020
	(0.013)	(0.019)	(0.020)	(0.013)	(0.012)	(0.012)	(0.013)
Control Mean	0.667	0.531	0.466	0.559	0.618	0.563	0.617
Observations	4,154	2,020	1,804	4,548	4,430	4,864	4,114

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Each column restricts the sample to the subgroup in the column header. Each cell is from a separate regression. All models include fixed effects for college (strata), controls for demographic indicators (race, gender and age, Pell recipient status), GED indicator, and calculated math and English algorithm values. Pass rates are calculated as total college credits earned divided by total college credits attempted.

Table A.17. Selective Labels: Interaction Terms from Eq. [6]

	(1)	(2)
	Pass College-level English	Pass College-level Math
selected × age	-0.010	0.002
	(0.009)	(0.007)
selected × $\mathbf{1}[25 < \text{age}]$	0.134	-0.172
	(0.148)	(0.128)
selected × HS GPA	-0.021***	-0.008
	(0.008)	(0.005)
selected × ACPL Algebra	-0.020	-0.015
	(0.079)	(0.053)
selected × ACPL Arithmetic	-0.002	-0.052
	(0.052)	(0.078)
selected × ACPL Reading	-0.020	0.102**
	(0.068)	(0.050)
selected × ACPL Written	-0.226***	-0.021
	(0.081)	(0.056)
selected × ACPL Sentence	-0.008**	-0.006**
	(0.003)	(0.002)
selected × GED	0.134	0.264**
	(0.148)	(0.118)
Observations	903	1,560

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. This table reports the interaction terms for Eq. [6] in the text for passing the college-level English course and math course in Column (1) and (2), respectively. The estimation in each column is restricted to students in College 5 eligible for placement in English in Column (1) or Math in Column (2). See Table A.7 for descriptive statistics of College 5. All models include the reported interaction terms as well as each variable separably, and indicator variables for missing data and their interactions with $selected_i$.

Table A.18. Characteristics of Those Who Benefit from English Remedial Placement

	Do Not Benefit from English Remedial Placement	Benefit from English Remedial Placement	Difference	P-value	Obs.
Female	0.500	0.482	-0.018	0.354	11,883
Race					
White	0.434	0.364	-0.070	0.000	12,526
Asian	0.025	0.017	-0.008	0.100	12,526
Black	0.193	0.260	0.067	0.000	12,526
Hispanic	0.198	0.200	0.002	0.884	12,526
Other	0.051	0.042	-0.009	0.198	12,526
Age at entry	20.93	20.85	-0.081	0.702	12,526
Pell Grant recip.	0.453	0.516	0.063	0.001	12,526
TAP Grant recip.	0.309	0.332	0.022	0.203	12,526
GED recip.	0.070	0.045	-0.025	0.001	12,526
HS GPA (100 scale)	78.44	72.65	-5.794	0.000	7,857
ACCUPLACER Exam score					
Arithmetic	33.83	33.02	-0.805	0.416	10,187
Algebra	48.58	39.97	-8.609	0.000	10,187
College-level math	8.282	3.719	-4.563	0.000	3,655
Reading	57.96	59.95	1.982	0.039	12,526
Sentence skills	34.61	36.76	2.144	0.002	10,709
Written exam	3.879	4.171	0.292	0.000	10,968
Total Obs.	11,751	775			12,526

Notes: Sample includes any student who took the English placement exam at one of the seven study colleges and got a different placement from the ACCUPLACER and the algorithmic systems. Currently dual enrollment students and students who tested into ESL courses are excluded. The difference column shows estimates from a regression of the baseline characteristic shown on the lefthand side on a "benefits from remedial placement" indicator and strata fixed effects (indicators for each college). The causal forest model (Athey et al., 2019) was trained to predict the conditional average treatment effect of English remedial placement on the total number of credits earned. Results do not vary meaningfully from those obtained using the test set sample.

Table A.19. Characteristics of Those Who Benefit from Math Remedial Placement

	Do Not Benefit from Math Remedial Placement	Benefit from Math Remedial Placement	Difference	P-value	Obs.
Female	0.499	0.498	-0.001	0.938	11,883
Race					
White	0.430	0.426	-0.004	0.791	12,526
Asian	0.023	0.035	0.012	0.046	12,526
Black	0.199	0.177	-0.022	0.091	12,526
Hispanic	0.198	0.200	0.002	0.897	12,526
Other	0.050	0.053	0.003	0.721	12,526
Age at entry	20.99	20.40	-0.581	0.016	12,526
Pell Grant recip.	0.457	0.459	0.002	0.887	12,526
TAP Grant recip.	0.313	0.284	-0.029	0.054	12,526
GED recip.	0.072	0.035	-0.037	0.000	12,526
HS GPA (100 scale)	77.62	81.31	3.692	0.000	7,857
ACCUPLACER Exam score					
Arithmetic	34.22	30.54	-3.67	0.000	10,187
Algebra	46.42	59.74	13.33	0.000	10,187
College-level math	5.413	18.08	12.66	0.000	3,655
Reading	58.59	53.43	-5.16	0.000	12,526
Sentence skills	37.06	9.00	-28.06	0.000	10,709
Written exam	3.921	3.698	-0.223	0.004	10,968
Total Obs.	11,301	1,225			12,526

Notes: Sample includes any student who took the math placement exam at one of the seven study colleges and got a different placement from the ACCUPLACER and the algorithmic systems. Currently dual enrollment students and students who tested into ESL courses are excluded. The difference column shows estimates from a regression of the baseline characteristic shown on the lefthand side on a "benefits from remedial placement" indicator and strata fixed effects (indicators for each college). The causal forest model (Athey et al., 2019) was trained to predict the conditional average treatment effect of math remedial placement on the total number of credits earned. Results do not vary meaningfully from those obtained using the test set sample.

Table A.20. Impacts on Credits Attempted and Earned. Full Sample

Per-student outcomes	Control	Treatment	Difference
Remedial credits:			
Attempted	3.539	2.444	-1.095***
Earned	1.761	1.101	-0.660***
College credits in math/English:			
Attempted	6.889	7.260	0.371***
Earned	3.986	4.124	0.138*

Notes: Heteroskedasticity-robust standard errors shown in parentheses. *** p<0.01, ** p<0.05, * p<0.10. All models include fixed effects for each college, controls for demographic indicators including race, gender and age, Pell recipient status, GED indicator, and calculated math and English algorithm values.

Table A.21. Changes in Total Credits Attempted and Costs for Students

	Treatment Effect
Credits attempted relative to status quo	-0.724
Difference in credits paid by students	-$140

Source: Table A.20; authors' calculations. Cost figures rounded to nearest 10. The table covers remedial and college-level math and English credits; induced college-level credits in other subjects (Table A.9) are excluded.

Table A.22. Costs for Implementation and Operation of the Algorithmic Placement System

		Range Per College	
	Total (six colleges)	Lower Per-student Incremental Cost Bound	Upper Per-student Incremental Cost Bound
Students per semester	5,808	2,750	505
Total Placement Cost:			
Algorithm	$958,810	$268,890	$196,170
Test Score Placement	$174,240	$82,590	$15,150
New placement incremental cost:			
Per semester	$784,560	$186,300	$181,020
Per student	$140	$70	$360

Notes: 2016 dollars. Present values (discount = 3%). Rounded to $10. Ingredients information on full-time equivalents is from interviews with key personnel at six colleges. Lower and upper bounds represent highest and lowest per-student incremental costs across the six colleges. Cost data not available for one college. Costs amortized over cohorts. Student cohorts rounded to nearest 10. *Total placement cost* includes all costs to implement and administer the placement test; personnel (i.e., IT, program, senior/faculty, administrative support, and evaluator's time), fringe benefits, and overheads/facilities.
IT personnel salary data from https://www.cs.ny.gov/businesssuite/Compensation/Salary-Schedules/index.cfm?nu=PST&effdt=04/01/2015&archive=1&fullScreen.
Program personnel annual salary (step 4, grade 13) from https://www.suny.edu/media/suny/content-assets/documents/hr/UUP_2011-2017_ProfessionalSalarySchedule.pdf.
Senior/faculty midpoint MP-IV https://www.suny.edu/hr/compensation/salary/mc-salary-schedule/ https://www.cs.ny.gov/businesssuite/Compensation/Salary-Schedules/index.cfm?nu=CSA&effdt=04/01/2015&archive=1&fullScreen.
Evaluator's time estimated from timesheets. Fringe benefits uprated from ratio of fringe benefits to total salaries (IPEDS data (2013, 846 public community colleges). Overheads/facilities uprated from ratio of all other expenses to total salaries (IPEDS data (2013, 846 public community colleges). Cost to administer placement test from Rodríguez et al. (2014).
New placement incremental cost is the difference between the test-score placement system and the new, algorithmic placement system's total placement costs. More than two-thirds of the new placement incremental costs are implementation costs, and approximately 30% are operating costs ($40 per-student), which refer to running of new placement system after initial algorithm has been developed and tested.

Table A.23. Cost-Effectiveness Analysis: Social Perspective

Per-student Costs	Control	Treatment	Difference
Direct cost: Placement	$30	$170	$140
Indirect cost: Attempted remedial credits	$1,840	$1,270	-$570
Indirect cost: Attempted math and English college credits	$3,580	$3,770	$200
Total Cost	***$5,450***	***$5,220***	***-$230***
Earned math and English college credits	3.986	4.124	0.138
Cost per earned college credit	$1,370	$1,270	-$100

Source: From the authors' calculations. Cost figures rounded to nearest 10. The table covers remedial and college-level math and English credits; induced college-level credits in other subjects (Table A.9) are excluded.